

Name:

Form:

MYP 2

Year 8

Energy Production and Climate Change

Statement of Inquiry:	Exploring changes in the environment brought about by energy production and investigating the consequences on humankind .
Key Concept:	Change
Related Concepts:	Energy and Consequences



2022

8C MYP Command Terms

Term	Definition
Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.
Apply	Use knowledge and understanding in response to a given situation or real circumstances
Describe	Give a detailed account or picture of a situation, event, pattern or process
Design	Produce a plan, simulation or model
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence
Document	Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography
Evaluate	Make an appraisal by weighing up the strengths and limitations
Explain	Give a detailed account
Formulate	Express precisely and systematically the relevant concept(s) or argument(s)
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
Outline	Give a brief account
Present	Offer for display, observation, examination or consideration
Recall	Remember or recognize from prior learning experiences
Select	Choose from a list or group
Solve	Obtain the answer(s) using appropriate methods
State	Give a specific name, value or other brief answer without explanation or calculation
Suggest	Propose a solution, hypothesis or other possible answer
Summarize	Abstract a general theme or major point(s)

Glossary: Energy Production & Climate Change

Word	Meaning
atomic energy	Another name for nuclear energy.
cell	It contains a store of chemical energy that can produce electricity (the scientific name for a battery).
chemical energy	The kind of energy stored in chemicals. Food, fuel and cells (batteries) all contain chemical energy.
electrical energy	The kind of energy carried by electricity.
energy flow diagram (Sankey)	A way of showing energy changes as a flow chart.
gravitational	The kind of energy stored by anything that can fall down.
heat energy	The hotter something is the more heat energy it has.
joule (J)	The unit for measuring energy.
kilojoule (kJ)	There are 1000 joules (J) in 1 kilojoule (kJ).
kinetic energy	The kind of energy in moving things.
light energy	The kind of energy given out by light bulbs, candles, etc.
machine	Something that changes energy from one form to another.
nuclear energy	Energy stored inside atoms.
potential energy	The scientific word for 'stored' energy.
sound energy	The kind of energy that is made by anything that is making a noise.
strain energy	The kind of energy stored in stretched or squashed things which can change back to their original shapes.
transfer	When energy is changed from one form into another we say it is transferred.
coal	A fossil fuel formed from the remains of plants and trees.
fossil fuels	Coal, oil and natural gas – all fuels that were formed over millions of years from the remains of dead plants and animals.
oil	Fossil fuel formed from the remains of dead animals that lived in the sea.
natural gas	Fossil fuel formed from the remains of animals that lived in the sea.
non-renewable energy resource	Any energy resource that will run out and we cannot renew our supplies of it (e.g. oil).
solar panels	Flat plates that use the Sun's energy to heat water.

Word	Meaning
Non-renewable energy resource	An energy resource that is NOT easily replaced and will run out (e.g. oil).
hydroelectric power	Making electricity by letting falling water (usually from a reservoir) turn turbines and generators.
geothermal power	Making electricity using heat from hot rocks underground.
biomass	Any fuel that comes from plants, animals, or their wastes (e.g. wood, methane from rotting plants, etc.).
solar cells	Flat plates that convert light energy into electrical energy.
wind turbine	A kind of windmill that generates electricity using energy from the wind.
turbine	The machine in a power station that is pushed round by water or steam and turns the generator.
nuclear power	Making electricity by using the nuclear energy stored inside uranium.
tidal power	Making electricity using the moving (kinetic) energy from the tides.
conduction	The way heat travels through solids.
insulator	A material which does not let energy flow through it easily.
absorb	To take in energy.
convection	The transfer of heat in fluids.
convection current	A current created by heat causing changes in the density of a fluid.
emit	To give out (radiate) energy.
infrared radiation	A type of wave in the electromagnetic spectrum. It can travel through transparent things and a vacuum (empty space).
medium	Any substance (that a wave is travelling through).
transfer	When energy is changed from one form into another we say it is transferred.
ammeter	Measures how much electricity is flowing around a circuit.
current	The flow of electrons around a circuit.
in parallel	When the current divides, a part going through each component, then joins up to complete the circuit.
voltage	A measure of how much energy is transferred by electricity.
voltmeter	A component that measures voltage.
volt (V)	The unit for voltage.
rechargeable resource	Cells that can have more energy stored in them after they have been used are said to be rechargeable.

8P 1a Energy – What I already know!

Please answer the questions below. If you are not sure of the answer just do your best and write what you think you know about the question. You will keep this questionnaire and it will help you see how much you have learnt over this topic.

Question	Answer at the start of the energy topic	Answer at the end of the energy topic
How would you describe the term energy?	Energy is the ability to make something happen or change - it is needed to do work. Energy is the ability to do work or cause change. It cannot be created or destroyed, only transferred, stored or dissipated.	Energy is the ability to make something happen or change - it is needed to do work.
List as many types of energy as you can think of.	Light, heat (thermal), sound, electrical, kinetic (movement), gravitational potential, strain (elastic), chemical, nuclear.	Light, heat (thermal), sound, electrical, kinetic (movement), gravitational potential, strain (elastic), chemical, nuclear.
What is electricity?	Electricity is the flow of electric charge (electrons) around a circuit, carrying electrical energy. Electricity is the flow of electric charge (electrons) around a complete circuit; the current transfers energy from a cell or power supply to the components.	Electricity is the flow of electric charge (electrons) around a circuit, carrying electrical energy.
What does the term conduction mean?	Conduction is the transfer of heat through a solid by particles vibrating and passing energy along to their neighbours.	Conduction is the transfer of heat through a solid by particles vibrating and passing energy along to their neighbours.
Conduction is the transfer of heat through a substance by particles vibrating and passing energy to neighbouring particles; it		

What are fossil fuels?	Coal, oil and natural gas - fuels formed over millions of years from the remains of dead plants and animals.	Coal, oil and natural gas - fuels formed over millions of years from the remains of dead plants and animals.
What are renewable and non-renewable resources?	Renewable resources (e.g. wind, solar, hydroelectric) will not run out as they can be replaced; non-renewable resources (e.g. coal, oil, gas) will run out and cannot be replaced.	Renewable resources (e.g. wind, solar, hydroelectric) will not run out as they can be replaced; non-renewable resources (e.g. coal, oil, gas) will run out and cannot be replaced.
What is climate change?	A long-term change in the Earth's climate, mainly caused by burning fossil fuels which release greenhouse gases such as carbon dioxide.	A long-term change in the Earth's climate, mainly caused by burning fossil fuels which release greenhouse gases such as carbon dioxide.

8P 1b Introduction to Energy Types

Introduction:

- **ENERGY** is

the ability to do work (the ability to make something happen or change)

never created or destroyed, only transferred from one type to another

measured in joules (J)

- The unit for energy is

the joule (J)

- There are **nine types of energy**.

List the nine types of energy below.

light energy

heat (thermal) energy

sound energy

electrical energy

kinetic (movement) energy

gravitational potential energy

strain (elastic potential) energy

chemical energy

nuclear (atomic) energy

- These different types of energy can be categorized into two main forms:

- **Kinetic** energy – ‘movement’ energy (‘energies in action’)
- **Potential** energy – ‘stored’ energy

Can you identify and highlight the types of energy that fall under the **kinetic energy** category and also the ones that fall under the **potential energy** category in your list?

What **type(s) of energy** does the following have?

1. Moving car

kinetic (movement) energy

2. Battery

chemical energy

3. Diver at the top of the cliff

gravitational potential energy

4. A torch switched on

light energy (and heat/thermal energy)

5. Stretched spring

strain (elastic potential) energy

6. Fire

heat (thermal) energy and light energy

7. Loudspeaker playing music

sound energy (and kinetic energy of the cone)

8. Nuclear power plant

nuclear energy (transferred to heat then electrical energy)

9. Book on the edge of a table

gravitational potential energy

10. Wind turbine

kinetic energy of the moving air, transferred to electrical energy

For each type of energy, can you think of your own example(s)? Write them down below.

Light energy - light from the Sun, a torch or a lamp

Heat (thermal) energy - a hot drink, a radiator or a Bunsen flame

Sound energy - a ringing bell or a guitar string

Electrical energy - the mains electricity that lights a bulb

Kinetic energy - a moving car or a running student

Gravitational potential energy - a book on a shelf

Strain energy - a stretched rubber band or a squashed spring

Chemical energy - food, a battery or petrol

Nuclear energy - energy stored inside the atoms of uranium

Moving air (wind) has kinetic energy - it turns a wind turbine

8P 2a Energy Circus

Law of Conservation of Energy

In 1842, Julius Robert Mayer discovered the Law of Conservation of Energy.

This important law states that energy can be transferred from one type to another, but it can never be made or destroyed.

So this means that the total amount of energy in the Universe stays the same!

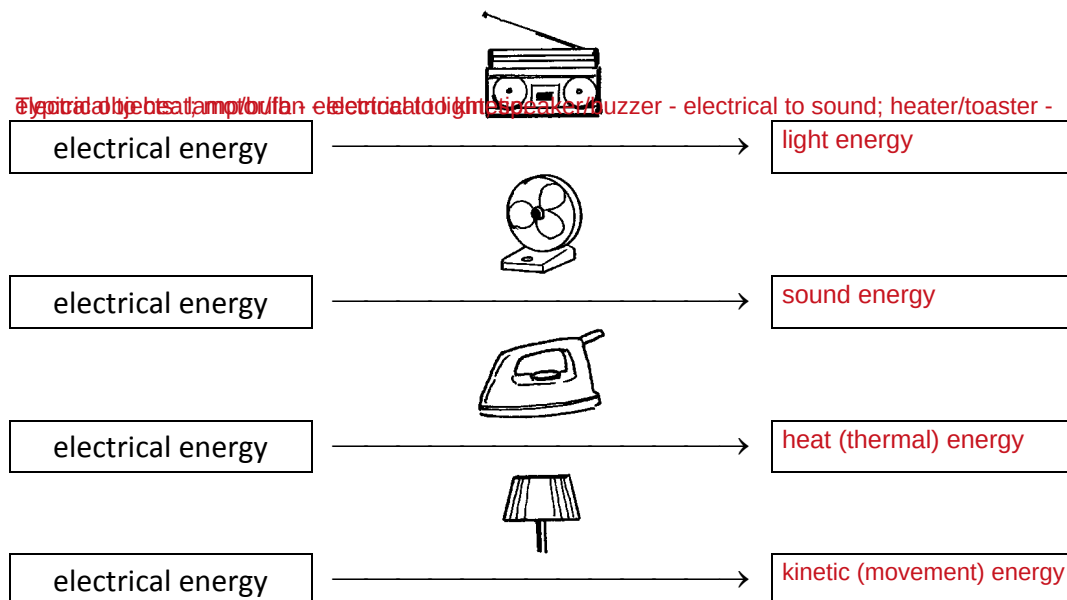
Energy Transfers

We can write energy flow diagrams to show the energy transfers that occur in objects and also in given situations. For instance,

- Objects:

The pictures below show some objects that need electrical energy to work.

Write the useful energy each object produces in the boxes.



Notice that the electrical energy is transferred to a different type of energy. The transfer of energy or flow of energy is illustrated by the arrow.

- Situation:

When a car brakes...

Chemical —————→ Kinetic —————→ Heat/thermal

When a rocket launches...

Chemical —————→ Kinetic + Sound + Heat —————→ Gravitational

Practical - Energy Circus

In this practical, you have about 3 mins to think and write down the **energy transfer(s)** for each of the stations set up around the lab. Please write down the energy transfers below.

- 1) Solar Cell powered light or fan

Light energy to electrical energy (in the solar cell) to light energy (in the lamp) - or to kinetic energy if the fan spins.

- 2) Pendulum set up and/or yoyo

Gravitational potential energy to kinetic energy (and back again), plus a little wasted heat and sound.

- 3) Wind-up toy. Strain (elastic potential) energy in the wound-up spring to kinetic energy of the toy, plus lost sound and heat.

- 4) "Bungee sheep" or similar.

Gravitational potential energy to kinetic energy, then to strain (elastic potential) energy in the bungee as it stretches, then back to kinetic energy.

5) Electric bell Electrical energy to kinetic (movement) energy of the hammer and sound energy

6) Electric light Electrical energy to light energy and wasted heat (thermal) energy

7) Simple pendulum Gravitational potential energy to kinetic energy and back to gravitational potential energy (a little heat and sound wasted)

8) Water turbine powering a LED

Gravitational potential and kinetic energy of the water to kinetic energy of the turbine to electrical energy to light energy in the LED

9) Signal Generator and Speaker Electrical energy to sound energy (and wasted heat energy)

10) Hairdryer Electrical energy to heat (thermal) energy, kinetic (movement) energy of the air and some sound energy

8P 2b Energy Circus

Teacher Demo – Steam Engine



What are the energy transfers for this steam engine?

Chemical energy stored in the fuel is transferred to heat (thermal) energy when the fuel burns

The heat energy boils water, giving the steam kinetic energy and pushing pressure

The moving steam is transferred to kinetic energy of the engine's moving parts

Some energy is transferred to sound energy

Some energy is wasted as heat (thermal) energy to the surroundings

Overall: chemical energy to heat energy to kinetic energy (plus wasted sound and heat)

Sankey Diagram

- We can represent **energy transfers** in a Sankey diagram.
- A Sankey diagram shows:
 - **energy input**
 - **useful output**
 - **energy wasted (non-useful).**

It's called a Sankey diagram as it was named after the engineer Captain Henry Sankey. He created a diagram showing energy flow in a real steam engine compared to a 'perfect' engine in 1898.



Example: A **filament bulb** transfers electrical energy into heat and light energy.

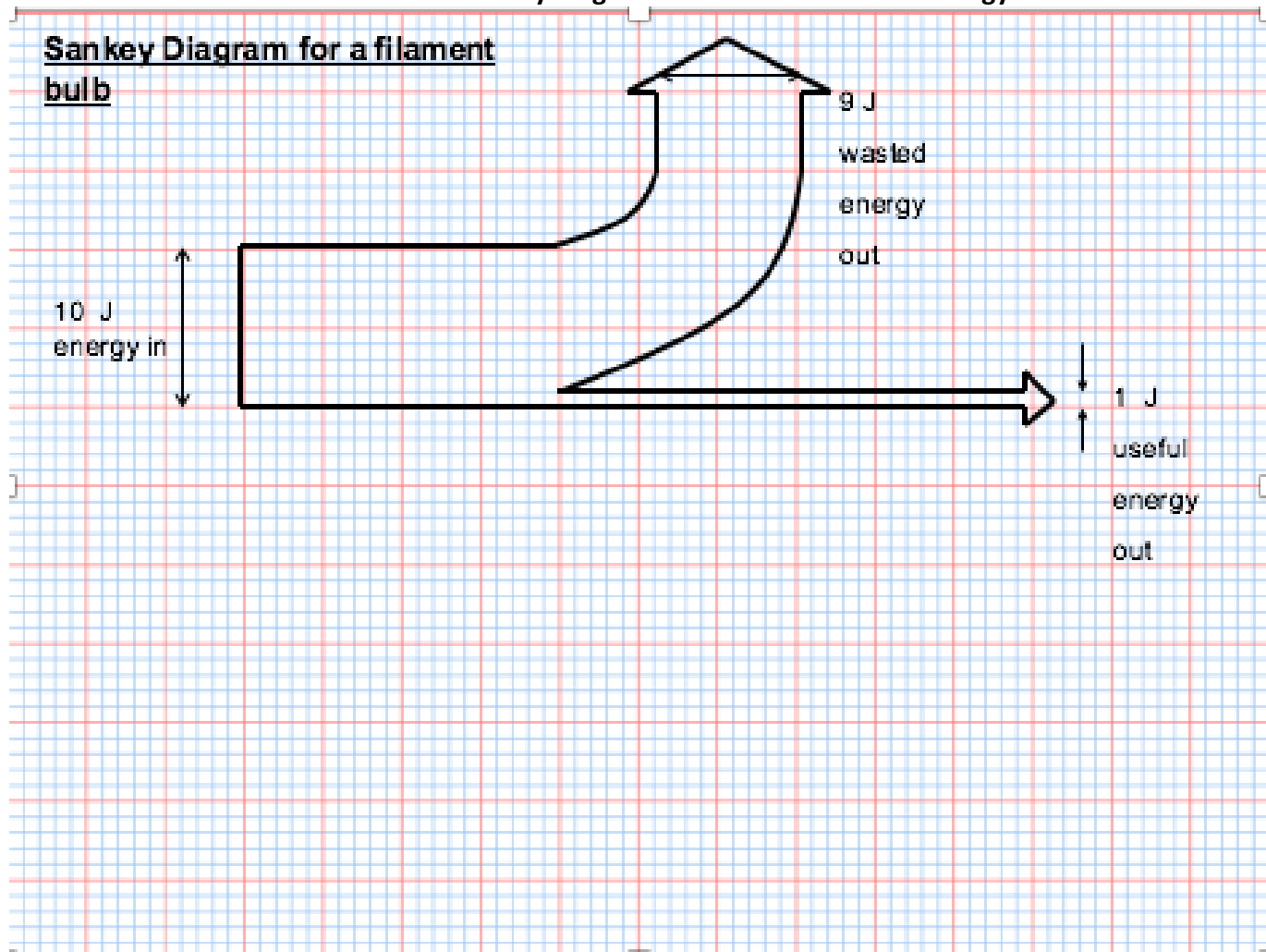
What is useful energy in this example?

The useful energy is the light energy given out by the bulb - that is what the bulb is used for.

What is wasted energy in this example?

The wasted energy is the heat (thermal) energy given out - it is not needed to light the room.

Note that the width of an arrow in a Sankey diagram shows the size of the energy flow.



Use the squares on the previous page to help you draw a Sankey diagram for an **energy efficient bulb**, which has the following:

- Energy input 10J
- Wasted energy 6J
- Useful energy 4J

Calculate the efficiency for the filament bulb and the energy efficient bulb using:

$$\text{Efficiency (\%)} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

Energy efficient bulb: Efficiency = useful output / total input x 100 = 4 J / 10 J x 100 = 40%

Filament bulb: read the useful light energy and the total input energy from the Sankey diagram on page 13, then use

Efficiency (%) = useful energy output / total energy input x 100. Wasted heat energy = total input - useful output.

Efficiency (%) = (useful energy output ÷ total energy input) × 100

Useful energy output = the energy transferred to the wanted (useful) form, in joules.

Total energy input = all the energy supplied to the device, in the same unit (J or kJ).

Multiply the fraction by 100 to make it a percentage; efficiency is never more than 100%.

Worked example: 40 J useful output from 100 J input gives (40 ÷ 100) × 100 = 40% efficient.

Wasted energy = total input - useful output; the more energy wasted, the lower the efficiency.

Wasted energy is usually transferred to the surroundings as heat (thermal energy), and sometimes as sound.

Both energy values must be in the same unit before dividing; the answer has no unit because it is a percentage.

Efficiency can also be given as a decimal or fraction: 0.4 = 4/10 = 40%.

Extension

1. An energy efficient bulb transfers 40% of the energy as useful light energy.

- a. How much energy is transferred as heat?

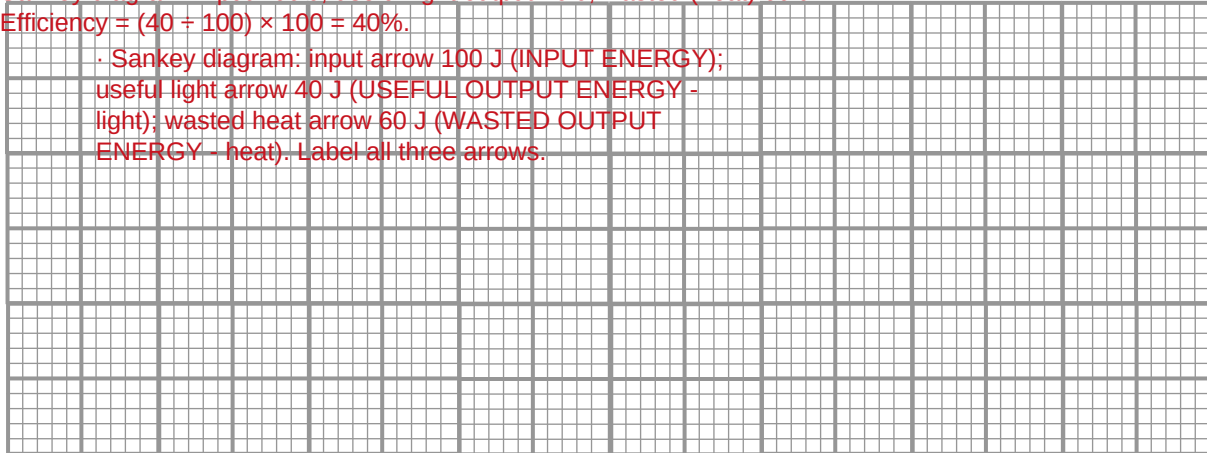
60% of the energy is transferred as heat (60 J out of every 100 J)

- b. Draw a Sankey Diagram in the space below

Sankey diagram: input 100 J, useful light output 40 J, wasted (heat) 60 J.

Efficiency = $(40 \div 100) \times 100 = 40\%$.

Sankey diagram: input arrow 100 J (INPUT ENERGY); useful light arrow 40 J (USEFUL OUTPUT ENERGY - light); wasted heat arrow 60 J (WASTED OUTPUT ENERGY - heat). Label all three arrows.



Remember to label it with: INPUT ENERGY, USEFUL OUTPUT ENERGY, and WASTED OUTPUT ENERGY

2. A team of scientists test an “old banger” car. They find out that only 10% of the energy is transferred as kinetic energy (useful) and 70% is transferred as heat energy (wasted).

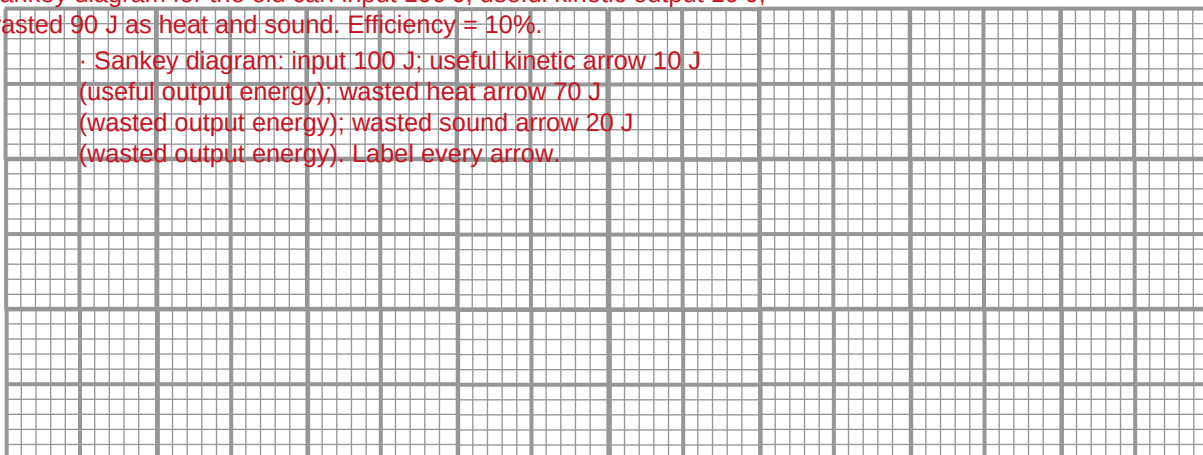
- a. How much energy is wasted as sound?

20% of the energy is wasted as sound (100% - 10% - 70% = 20%)

- b. Draw a Sankey Diagram in the space below

Sankey diagram for the old car: input 100 J, useful kinetic output 10 J, wasted 90 J as heat and sound. Efficiency = 10%.

Sankey diagram: input 100 J; useful kinetic arrow 10 J (useful output energy); wasted heat arrow 70 J (wasted output energy); wasted sound arrow 20 J (wasted output energy). Label every arrow.



Label the arrows with input energy, useful output energy (kinetic), wasted output energy (heat), wasted output energy (sound)



3. A team of scientists test a brand new hybrid car. They find out that 40% of the energy is transferred as kinetic energy (useful) and 55% is transferred as heat energy (wasted).

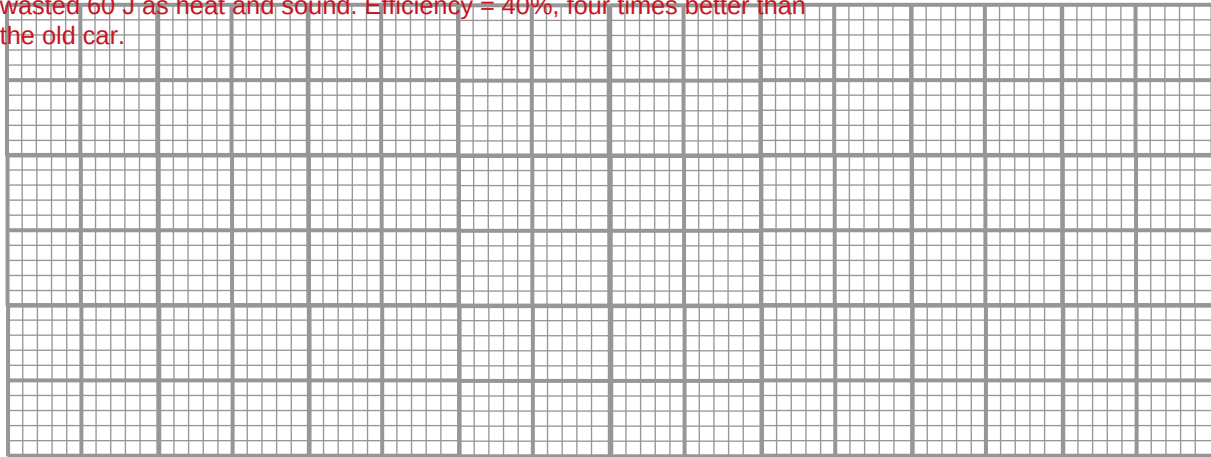


- a. How much energy is wasted as sound?

5% of the energy is wasted as sound (100% - 40% - 55% = 5%)

- b. Draw a Sankey Diagram in the space below

Sankey diagram for the hybrid: input 100 J, useful kinetic output 40 J, wasted 60 J as heat and sound. Efficiency = 40%, four times better than the old car.



Remember to label each of the arrows!!!

· Sankey diagram: input 100 J; useful kinetic arrow 40 J (useful output energy); wasted heat arrow 55 J; wasted sound arrow 5 J. Label every arrow.

4. Explain why driving a newer car is 'greener' than driving a very old car.

A newer car transfers a bigger fraction of its energy to useful kinetic energy, so less is wasted as heat and sound.

It therefore burns less fuel for the same journey, releasing less carbon dioxide, so it adds less to climate change

(it is more efficient, so it is greener),

A newer car is more efficient, so a larger proportion of the energy in the fuel is transferred as useful kinetic energy.

Less energy is wasted as heat and sound, so the car needs less fuel to travel the same distance.....

Burning less fuel releases less carbon dioxide (a greenhouse gas), so it adds less to global warming and climate change.

8P 3a Heat

Introduction:

Different types of energy can be converted into **thermal energy** (heat).

- **HEAT** can be transferred from one place to another.
- Heat always travels FROM hotter places/substances TO cooler places/substances.
- **Thermal or heat energy** is produced when an increase in temperature causes atoms and molecules to move faster and collide with each other.

Make a sketch of particles arranged in a solid, liquid and gas.

SOLID	LIQUID	GAS
· SOLID: particles drawn close together in a regular pattern, touching, just vibrating in fixed positions.	· LIQUID: particles drawn close together but randomly arranged, able to slide past each other.	· GAS: particles drawn far apart and randomly arranged, moving quickly in all directions.

Can you predict and sketch what the particles would look like when heated to a warm temperature? Use red and blue to represent hot and cold particles respectively.

SOLID	LIQUID	GAS
· Heated SOLID: same close-packed regular arrangement but the particles vibrate faster - draw most particles red (hot) to show more energy.	· Heated LIQUID: particles move faster and spread slightly further apart - draw most particles red (hot).	· Heated GAS: particles move even faster and are further apart than in the cold gas - draw most particles red (hot).



8P 3b Heat Transfer in Solids: Conduction

Introduction:

Heat transfer can happen by conduction. Conduction typically occurs in solids and it happens because of the vibration of atoms.

Materials which let heat travel along them are good conductors. Some materials are better conductors than others.

Practical: Conduction Experiment

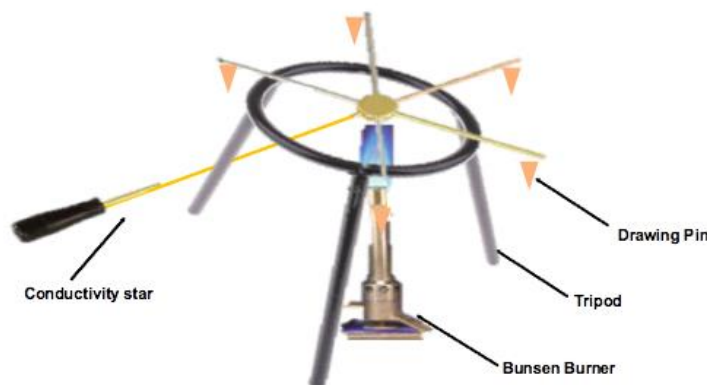
Your teacher will demonstrate this practical. You must record your results.

SAFETY:

- You must wear SAFETY GLASSES and if you have long hair, please tie it back.
- The rods will get very hot! Let them cool down before you touch them.

Apparatus:

- Bunsen burner
- Lighter
- Tripod stand
- Conductivity star (copper, aluminium, brass, steel and iron)
- Vaseline
- 5 drawing pins
- Stopwatch



Method

1. Place the conductivity star carefully onto the tripod stand.
1. Attach a drawing pin to the outer end of each metal using a small blob of Vaseline.
2. Heat the middle of the conductivity star with a blue Bunsen flame.
3. Record how long it takes for each metal to lose its drawing pin using the stopwatch.

Results

Metal	Time taken for the drawing pin to fall (seconds)
Copper	Approx. 5 s (fastest metal)
Aluminium	Approx. 10 s
Brass	Approx. 22 s
Steel	Approx. 40 s (slowest metal)
Iron	Approx. 35 s

Analysis

1. Explain why the drawing pin/ piece of metal falls.

The Bunsen flame heats the metal rod. The heat energy is conducted along the metal: its particles gain energy, vibrate

faster and pass the energy to the neighbouring particles. When the heat reaches the outer end the Vaseline melts, so the drawing pin is no longer stuck on and falls off.

Heat energy from the Bunsen flame is conducted along the metal rod to its outer end.

When the heat reaches the outer end the Vaseline melts, so the drawing pin is no longer stuck and falls off.

2. Rank the metals in order of conductivity.

Copper (best conductor), then aluminium, then brass, then iron, then steel (poorest conductor).

The metal whose drawing pin falls first is the best conductor of heat, because the heat travelled along it fastest.

Rank by the time taken for the pin to fall, shortest first (best conductor first), e.g. copper, aluminium, brass, iron/steel.

3. Explain why this experiment would not work if we used plastic or glass.

Plastic and glass are insulators - poor conductors of heat. Heat energy cannot travel along them, so the Vaseline at the far end would not melt and the drawing pins would not fall.

Plastic and glass are thermal insulators · they have no free electrons and conduct heat very poorly.

So heat would not travel along the rod to the outer end and the Vaseline would not melt · the pins would not fall.

8P 3c Heat Transfer in Liquids and Gases: Convection

Introduction:

Heat transfer can happen by convection. Convection can only happen in fluids (liquids and gases) because the molecules are free to move.

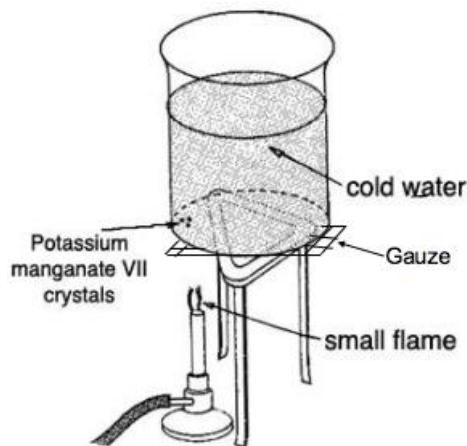
Practical: Convection current in a beaker

SAFETY:

- You must wear SAFETY GLASSES and if you have long hair, please tie it back.
- Potassium permanganate, also known as potassium manganate VII, is harmful if swallowed. It also stains skin and will leave permanent stains on clothing. So please take extra care when handling this chemical.

Apparatus

- Bunsen burner
- Lighter
- Tripod
- Gauze
- Beaker 600ml
- Glass tube
- Forceps
- Potassium permanganate crystals



Method

1. Fill a beaker with water.
2. As shown in the diagram, place the beaker of water on top of the gauze and tripod.
3. Then place a glass tube inside the beaker of water so that the end of the glass tube touches the inside corner of the beaker.
4. Using forceps, carefully place a potassium permanganate crystal through the glass tube so that the potassium permanganate crystal slides down to the corner of the beaker into the water.
5. Slowly remove the glass tube from the beaker. **DO NOT TOUCH OR MOVE THE BEAKER!**
6. Place the Bunsen burner directly underneath the crystal and heat it with a small flame.
7. Record your observations.

Observations

- Describe what you see.

When the potassium permanganate crystal is heated, a purple trail of colour rises from the crystal in the corner of the beaker.

The purple water moves up the side of the beaker above the flame, across the top of the water and down the other side,

The bottom spreads through the water and eventually all of the water in the beaker turns purple.

The moving purple water shows a convection current - the hot water is less dense so it rises and cooler water sinks to take its place.

The purple colour from the potassium permanganate rises from the crystal and spreads upward through the water,

showing convection currents: warm water rises, cooler water sinks.

- Sketch what you see. Use arrows to depict the direction in which the fluid moves.
 - Sketch: beaker of water with a loop of arrows - arrows going UP on the side above the flame, ACROSS the surface, DOWN the opposite side and back along the bottom (a convection current), with purple colour marked near the crystal.

8P 3d Heat Transfer: Radiation

Research about heat transfer by radiation.

➤ Questions to answer or think about:

What is heat transfer by radiation mean?

What's the difference between this type of heat transfer compared with conduction and convection?

Can you give examples of heat transfer by radiation?

Radiation is the transfer of heat energy by infrared waves (part of the electromagnetic spectrum).

Radiation is the transfer of heat by infrared (electromagnetic) waves. It needs no particles, so it can travel through a vacuum, e.g. from the Sun to Earth.

It can travel through transparent things and through a vacuum (empty space) - no particles are needed.

Conduction and convection both need particles (a medium) to transfer heat, whereas radiation does not.

Conduction happens in solids and convection only in fluids (liquids and gases); radiation can happen anywhere.

Examples: heat from the Sun reaching the Earth, warmth felt from a fire or a radiator, heat from a hot light bulb, heat from a toaster element.

Heat from the Sun reaching the Earth through the vacuum of space.

Heat from a fire or burning embers felt on your skin without touching it.

Heat from an electric bar heater or radiator warming a room.

Heat from a hot light bulb or glowing filament felt by a hand held near it.

Heat felt above hot tarmac or a hot pavement on a sunny day.

Heat from an oven or toaster felt on your face and hands.

A black surface, such as a car bonnet or tarmac, warming up in the Sun.

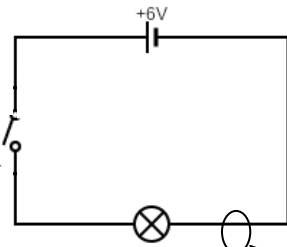
The warmth felt from a hot drink when your hand is held near the cup.

Infrared cameras and thermometers detecting the radiation given out by warm objects, people or animals.

8P 4a Introduction to Electric Circuits

What is Electricity? Electricity is the flow of electric charge (electrons) around a circuit; the current transfers energy from the cell or battery to the components.

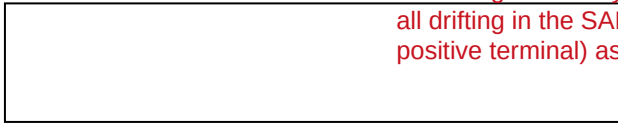
Here is a circuit diagram of a circuit containing a cell, switch and bulb:



Imagine you can see inside the wire. How does the electricity look as it flows through the wire?

Draw what you think the electricity looks like as it flows through the wire.

· Drawing: show tiny negatively charged particles (electrons) spaced along the wire, all drifting in the SAME direction (arrows from the negative terminal to the positive terminal) as they flow through the wire.



Enlarged View of Wire

Explain what you have drawn above. Try to reference some scientific concepts and ideas.

I have drawn the electrons - tiny negative particles that are already inside the metal wire.

The cell pushes the electrons around the circuit because it provides a voltage (potential difference).

The electrons drift around the whole circuit, transferring electrical energy to the bulb, which lights up.

This flow of electrons around the circuit is called the current, measured in amperes (A) with an ammeter.

Modelling:

Your teacher will now show you a few ways to model the flow of electricity through a wire, and some simulations. Please add some notes and diagrams to describe them in the space below.

Here are some words that you may wish to include in your notes:

· Model notes: the electrons in a wire behave like a moving queue or like a row of dominoes - the cell pushes the electrons and they all start moving, so the current flows all around the circuit at once. Use the words electrons, current, cell, circuit, energy, flow.



8P 4b Investigating Voltage and Current

Research Question:

To investigate the relationship between the current in a resistor and the voltage across it.

Hypothesis:

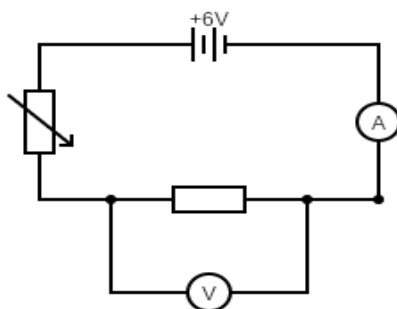
As the current in the resistor increases, the voltage across it will increase/decrease.

I think this because...

A larger current means more charge (electrons) flows through the resistor every second, so more energy is transferred by that charge.

If the resistance stays the same, more current must give a bigger voltage, so $V = I \times R$ shows the voltage will increase.

Apparatus:



Method:

1. Set up the circuit as above.
2. Set the supply voltage to 6V by adjusting the dial on the power pack.
3. Set the current to the lowest value by adjusting the variable resistor. This can be identified by looking at the ammeter.
4. Measure the corresponding voltage by reading the voltmeter.
5. Repeat steps 3 and 4 for up to 7 values if the circuit setup allowed in increments of 0.1A.

Results:

Take initial current reading then take readings in increments of 0.1A and record in the following table.

Current (A)	Voltage 1 (V)	Voltage 2 (V)	Average Voltage (V)
0.1	1.0	1.0	1.0
0.2	2.1	1.9	2.0
0.3	3.0	3.0	3.0
0.4	4.1	3.9	4.0
0.5	5.0	5.0	5.0
0.6	5.9	6.1	6.0

Graph:

Construct a graph to show the relationship between the voltage across the resistor and the current in the resistor. This should be done on graph paper and the following *criteria addressed*:

- *Title of graph included*
- *Label on x and y axis with correct units*
- *Scales should be appropriate. Equal divisions and occupy minimum 2/3 to ¾ of axis*
- *Points plots accurately*
- *Add a curve/straight line of best fit*

Conclusion:

1. What is the relationship between the current in a resistor and the voltage across it?

The voltage across the resistor is directly proportional to the current in it.

As the current increases the voltage increases in proportion (the graph is a straight line through the origin), because $V = I \times R$.

MYP 2 Science Energy Production & Climate Change Year 8

2. Does your conclusion agree with your hypothesis? If it does not, try to suggest reasons why.

Yes, my conclusion agrees with my hypothesis: as the current in the resistor increased, the voltage across it also increased.

The points on my graph formed a straight line through the origin, showing the two values rise together.

3. Explain your conclusion using scientific principles and ideas.

Current is the flow of electric charge (electrons) around a circuit, measured in amperes (A).

Voltage (potential difference) is the energy transferred by each unit of charge across the resistor.

If the current doubles, twice as much charge passes each second, so twice as much energy is transferred, so the voltage doubles.

This is Ohm's Law: voltage = current x resistance ($V = I \times R$), and the resistance of the resistor is constant.

Evaluation:

1. Suggest ways in which the accuracy, precision and reliability of the experiment can be improved.

Take repeat readings for each current value and calculate an average - this improves reliability and precision.

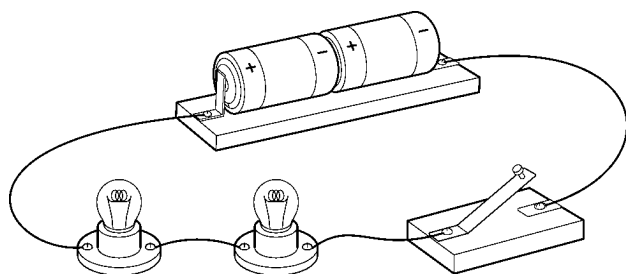
Use digital meters with a fine scale and read them at eye level to reduce parallax error and make the readings more accurate.

Keep the temperature of the resistor constant (switch off the circuit between readings) because its resistance changes when it heats up.

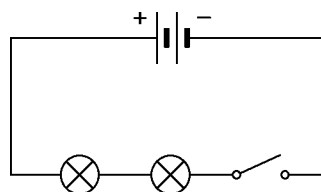
Check that the ammeter and voltmeter read zero before starting and use the same circuit and equipment each time.

Plot a line of best fit through all the points and repeat any anomalous results to check them.

8P 4c Circuit Diagrams



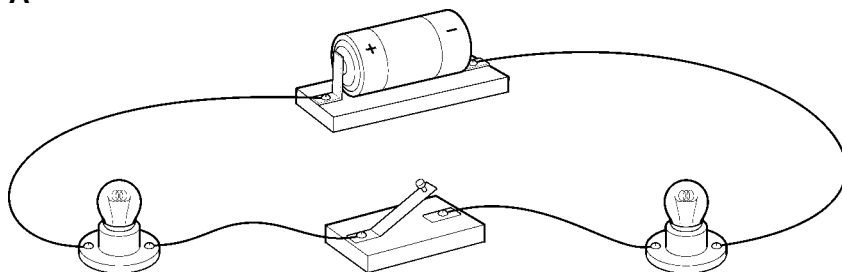
This is a circuit.



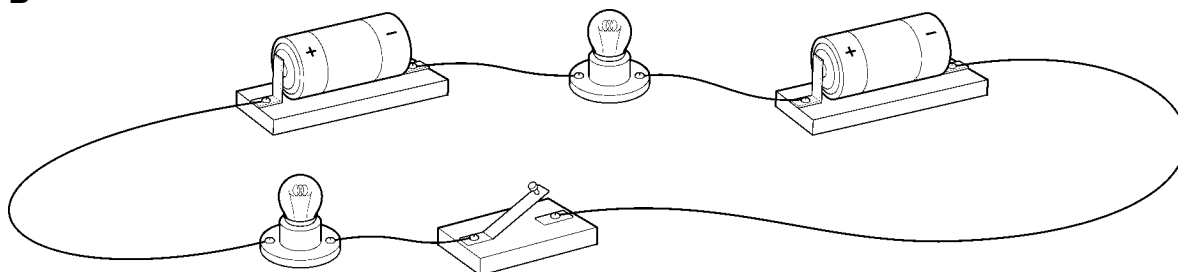
And this is the circuit diagram.

- 1 Make a neat copy of the circuit diagram above.
- 2 Draw circuit diagrams for these circuits. Make sure that you show the cells the right way around.

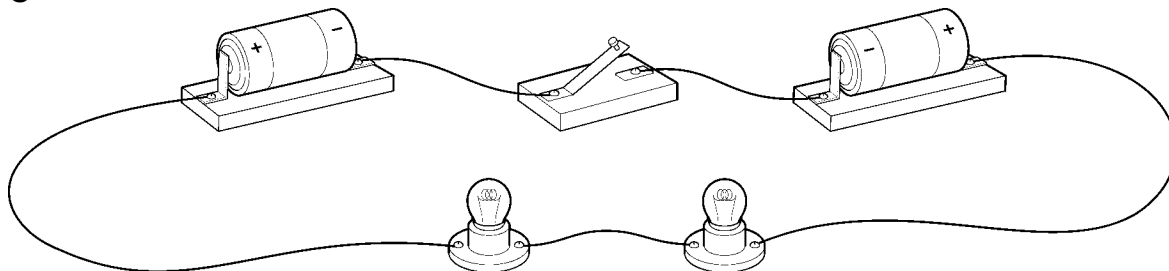
A



B



C



- 3 What is wrong with the cells in circuit C?
 · Neat copy of the circuit diagram: draw the cell (long line = positive terminal), the switch and the bulb joined in one loop using the correct circuit symbols and straight lines.

The two cells are connected the wrong way round - they face in opposite directions, so they oppose each other and their voltages cancel out, giving no (or very little) current.

· Circuit A diagram: draw the components in one series loop - cell, bulb and switch - using correct symbols and straight connecting wires.

8P 5a Series and Parallel Circuits

Introduction:

There are 2 main types of electric circuit:

1. Series
2. Parallel

You teacher will show you an example of each type of circuit and simulation. Draw the circuit diagram for each type of circuit in the space below.

SERIES

· **SERIES** circuit diagram: one single loop (e.g. cell, two bulbs and a switch) - the components are joined one after another so the same current flows through each.

PARALLEL

· **PARALLEL** circuit diagram: the circuit has branches (e.g. a cell with two bulbs in separate parallel branches) so the current divides between the branches and joins up again.

Describe the difference between a series and parallel circuit. Include diagrams if it helps.

In a series circuit all the components are joined in one single loop, so the same current flows through every component.

In a parallel circuit the current divides into separate branches, so part of the current goes through each component and then joins up again.

Two bulbs in series are dimmer and each has a smaller voltage, while two bulbs in parallel are brighter and each gets the full voltage.

8P 5b Current in Series Circuits

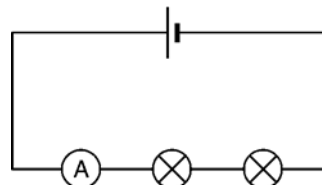
Apparatus:

Cells, Bulbs, Ammeter, Connecting wires

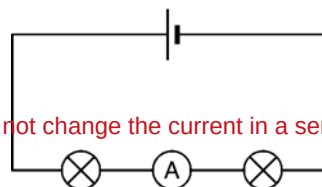
Method:

1. Build this circuit. Show it to your teacher before you switch on. Write down the reading on the ammeter.

Ammeter reading written down, e.g. 0.30 A. The current is the same at every point in a series circuit.

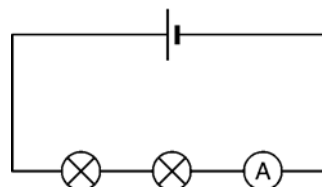


2. Now build this circuit. Write down the reading on the ammeter. Reading written down, e.g. 0.30 A - moving the ammeter does not change the current in a series circuit.



3. Build this circuit and write down the ammeter reading.

Reading written down, e.g. 0.30 A - the ammeter gives the same value wherever it is placed.



4. Now attach another cell into your circuit. Write down the new current value. Does the current increase or decrease? What happens to the brightness of the bulbs?

The current increases when another cell is added and the bulbs become brighter.

5. Finally attach a third bulb in series. Write down the current. Does adding an extra bulb increase or decrease the current? What does this do to the brightness of the bulbs?

Adding a third bulb in series decreases the current and makes the bulbs dimmer.

Conclusion:

1. What did you find out about the current in a series circuit?

The current is the same at every point in a series circuit.

The ammeter reading did not change as it was moved to different positions.

2. What effect does increasing the number of cells have on the current? Try to explain this.

Adding more cells increases the current.

More cells give a bigger push (higher voltage), so more current flows.

3. What effect does increasing the number of bulbs have on the current? Try to explain.

Adding more bulbs decreases the current.

Each extra bulb adds resistance, so less current flows around the circuit.

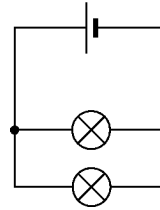
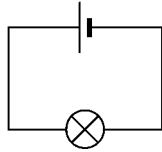
8P 5c Current in Series and Parallel

Apparatus:

Cells, Bulbs, Ammeter, Connecting wires

A. Bright Bulbs:

1. Build a simple series circuit below. Add an extra cell. Then add an extra bulb.
2. Build the simple parallel circuit below. Add an extra cell. Add an extra bulb in parallel.



Conclusions:

1. What happens to the brightness of bulbs in either circuit if you add more cells?

The bulbs get brighter in both circuits because the current increases.

2. What happens to the brightness of the bulbs in a **series** circuit if you add more bulbs?

In a series circuit the bulbs become dimmer - the current decreases.

3. What happens to the brightness of the bulbs in a **parallel** circuit if you add more bulbs?

In a parallel circuit the other bulbs stay the same brightness.

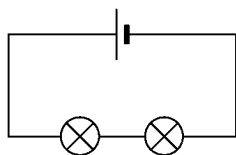
4. Can you explain why you think this happens?

In series, adding bulbs increases the resistance so the current falls.

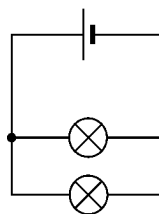
In parallel, each branch gets the full supply voltage and its own current.

So adding a bulb in parallel does not change the current in the other branches.

B. Missing Bulbs



This is a series circuit. If I take out one bulb it leaves a gap in the circuit. The other bulb goes off (goes off/stays on).



This is a parallel circuit. If I take out one bulb the other bulb stays on (goes off/stays on). There is still a complete circuit for the current to flow around through the other bulb.

Conclusion:

Which circuit is more useful and why?

A parallel circuit is more useful.

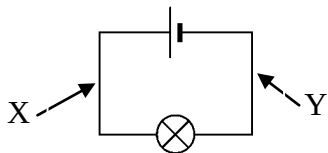
If one bulb breaks, the other bulbs keep working.

because there is still a complete circuit for the current to flow through.

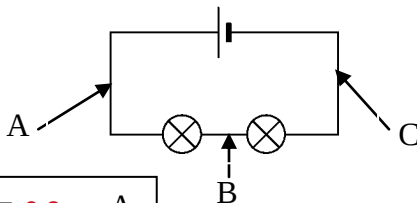
Each bulb can also be switched on and off separately.

C. Current in Series

Use an ammeter to measure the current in each of these circuits.



Current X = 0.3 A, Current Y = 0.3 A



Current A = 0.3 A

Current B = 0.3 A

Current C = 0.3 A

If you have time, put an extra cell into your series circuit and measure the current in positions A, B and C.

Conclusions:

1. What is the circuit symbol for an ammeter? A capital letter A inside a circle.
2. What unit is current measured in? Amperes - the amp, A.
3. What do you notice about the **current** in all parts of a series circuit?

The current is the same at every point in a series circuit.

4. What happens to the size of the current if more cells are added to the circuit?

The current increases, because the supply voltage is bigger.

5. What happens to the size of the current if more bulbs are added?

The current decreases, because the total resistance increases.

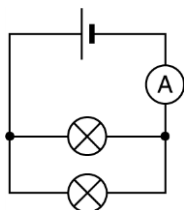
Extension: Can you explain any of this?

More cells push more current around. More bulbs add resistance, so the current falls.

D. Current in Parallel

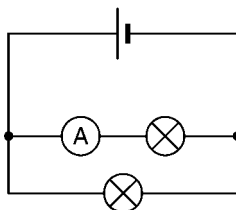
Construct these parallel circuits and measure the current in the positions indicated.

A



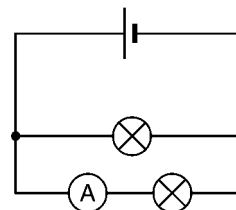
Current = 0.6 amps

B



Current = 0.3 amps

C



Current = 0.3 amps

Current A = 0.9 A

Current B = 0.45 A

Current C = 0.45 A

If you have time, add an extra cell to your circuit and measure the currents again.

If you have further time, add an extra bulb in parallel and measure the currents again. This time you will need to make a fourth measurement Current D.

Conclusions:

1. In a parallel circuit, what is the relationship between the current in the branches and the current in the main circuit?

The current in the main circuit equals the sum of the currents in the branches.

So the current divides at a junction and then rejoins afterwards.

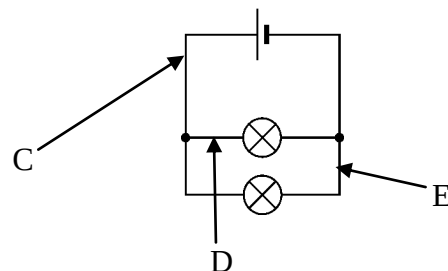
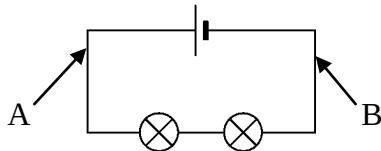
2. How does this help explain how we think current works in a parallel circuit?

Current is the flow of charge, and charge is conserved at a junction.

The charge flowing into a junction equals the charge flowing out of it.

so the branch currents add up to the current in the main circuit.

E. Switches in Series and Parallel circuits



Put some switches into the various positions shown in these series and parallel circuits.

Conclusion:

What happens when the switch is closed in a series circuit? In a parallel circuit, explain why the position of the switch is important?

In a series circuit closing the switch completes the circuit, so all the bulbs light.

In a parallel circuit a switch in the main circuit controls all the bulbs,

but a switch in one branch only controls the bulb in that branch.

Extension:

If you were wiring the lights in a house, would you use a series or parallel circuit? Why?

I would use a parallel circuit for the lights in a house.

Each bulb gets the full mains voltage, so they are all equally bright.

and if one bulb breaks or is switched off the others keep working.

Describe some advantages and disadvantages of series compared to parallel circuits.

Series: simple and cheap, one switch controls everything, but bulbs get dimmer as more are added and all go out if one breaks. Parallel: full brightness and independent switching, but needs more wire.

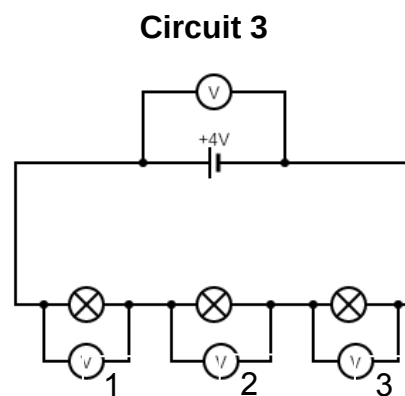
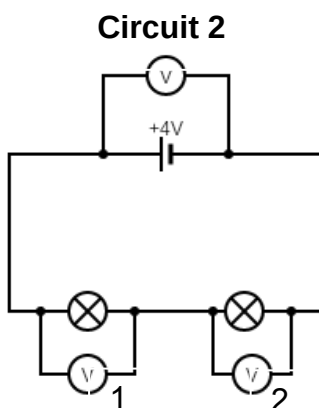
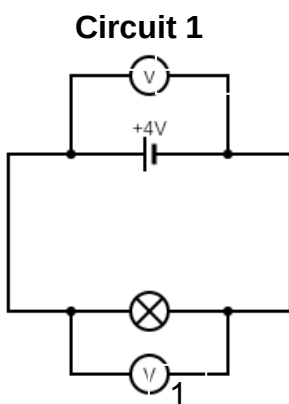
8P 5d Voltage in Series and Parallel Circuits

Apparatus:

Cells, Bulbs, Voltmeter, Connecting wires

A. Voltage in Series

Build the following circuits. Connect the voltmeter across each component in turn and record the voltage.



Results:

Circuit	Cell Voltage (V)	Lamp 1 Voltage (V)	Lamp 2 Voltage (V)	Lamp 3 Voltage (V)
1	1	1.5	1.5	n/a
2	2	1.5	0.75	0.75
3	3	1.5	0.5	0.5

Put an extra cell into your series circuit and measure the voltage across each. You will have to draw another table.

Conclusions:

1. What is the circuit symbol for a voltmeter? A capital letter V inside a circle.
2. What unit is voltage measured in? Volts, V.
3. What do you notice about the **voltage** across the cell and each of the components in the circuit?

The voltage of the cell equals the sum of the voltages across the components.

4. What happens to the voltage across each component if more cells are added to the circuit?

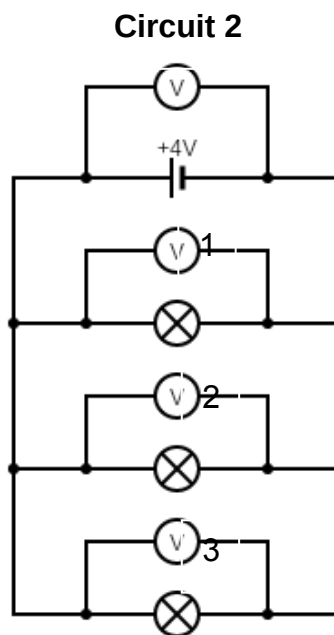
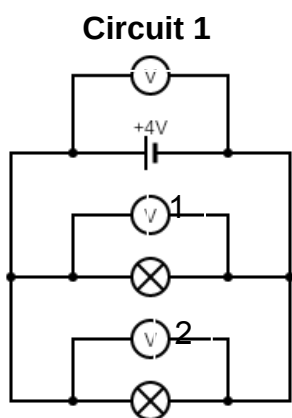
The voltage across each component increases, because the total supply voltage is greater.

5. What happens to the voltage across each component if more bulbs are added?

The voltage across each bulb decreases, because the supply voltage is shared between more components.

A. Voltage in Parallel

Build the following circuits. Connect the voltmeter across each component in turn and record the voltage.



Circuit	Cell Voltage (V)	Lamp 1 Voltage (V)	Lamp 2 Voltage (V)	Lamp 3 Voltage (V)
1	1	1.5	1.5	1.5
2	2	1.5	1.5	1.5

If you have time, add an extra cell to your circuit and measure the voltages again. You will have to draw another table.

Conclusion:

In a parallel circuit, what is the relationship between the voltage across each 'branch' and the voltage across the in the cell?

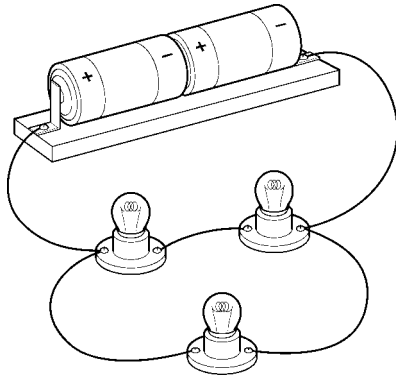
The voltage across each branch is the same as the voltage across the cell.

Every branch gets the full supply voltage, so the bulbs are all equally bright.

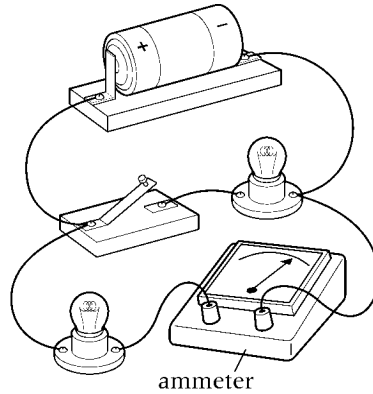
8P 5e - Review Circuit Diagrams and Current (optional)

1 Draw neat circuit diagrams to show these circuits in the space below.

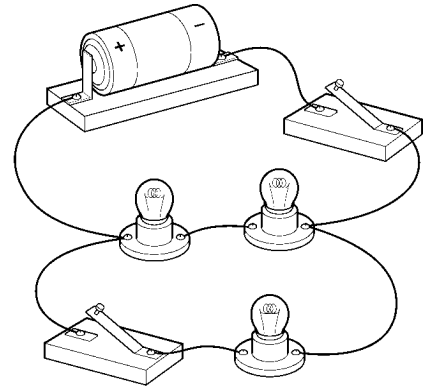
A



B



C



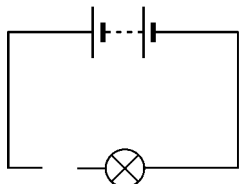
· Circuit A: redraw as a neat circuit diagram using correct symbols - cell, bulb(s), connecting wires - in a single loop (series).

· Circuit B: redraw as a neat circuit diagram using the correct symbols, showing each component clearly.

· Circuit C: redraw as a neat circuit diagram using the correct symbols, with straight lines and right-angled corners.

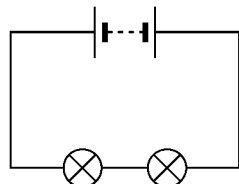
2 There is something wrong in each of these circuits. Write down what is wrong with each one.

D



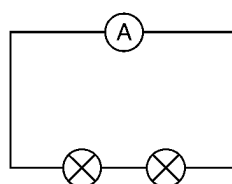
Circuit D: the bulb is short-circuited - a wire is connected across it so no current flows through the bulb.

E



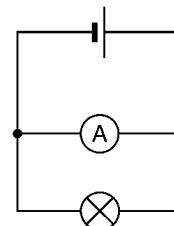
Circuit E: there is a gap in the circuit - it is not complete, so no current can flow.

F



Circuit F: there is no cell (power supply) in the circuit, so there is no push to make current flow.

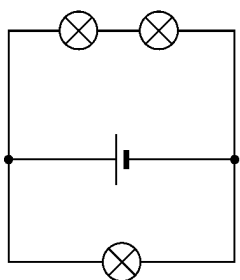
G



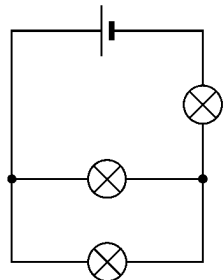
Circuit G: the cells are connected the wrong way round - they oppose each other so there is little or no current.

3 Which circuit is the odd one out? Explain your answer. (Tip: Redraw the circuits a different way)

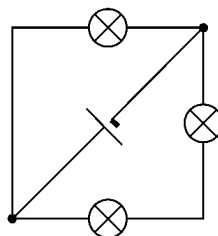
H



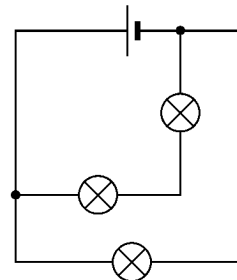
I



J

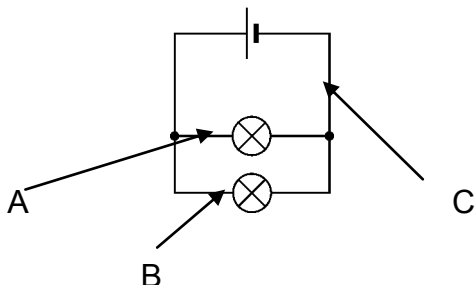


K



Circuit I is the odd one out. When H, J and K are redrawn a different way they are all the same circuit, but I is wired differently (the components are in the opposite order/it is a different circuit).

4.



If current A = 0.75 Amps and current B = 0.5 Amps, what should current C measure?

Current C should measure 1.25 A, because the current in the main circuit equals the sum of the branch currents ($0.75 + 0.5 = 1.25$ A).

8P 5f - Voltage and Current in Series and Parallel - Problems

Key Points:

Series Circuits:

- The supply voltage **divides** between the components in the circuit.
- The current is the **same** everywhere in the circuit.

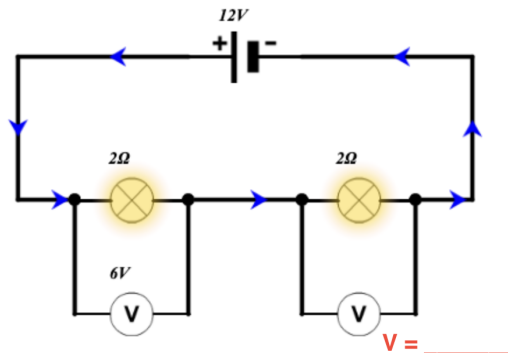
Parallel Circuits:

- The voltage across each branch is the **same** as the supply voltage.
- The current **divides** at each junction.

Example Problems:

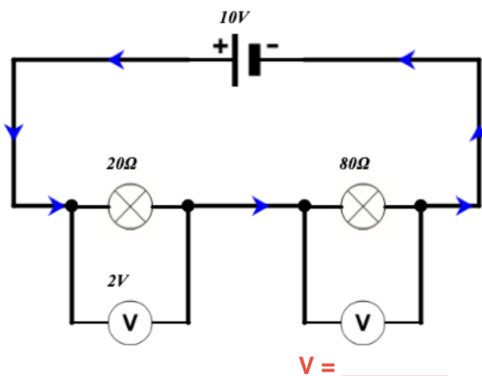
1. Find the unknown voltage for each circuit:

a.



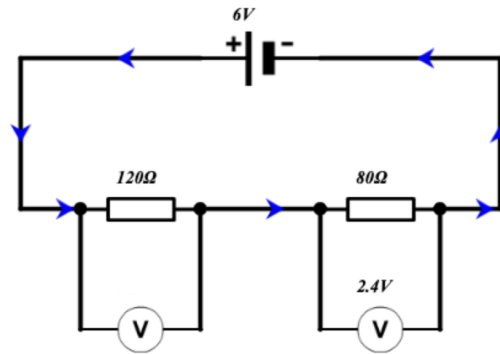
· Series circuit. The supply voltage divides between the components, so the unknown voltage = supply voltage - (sum of the other component voltages).

b.



· Parallel circuit. The voltage across each branch equals the supply voltage, so the unknown branch voltage = supply voltage (e.g. 6 V).

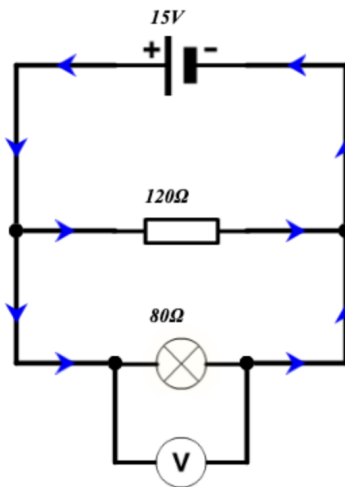
c.



V = _____

· Series circuit. Add up the voltages of the components to get the supply voltage, then subtract the known voltages to find the unknown one.

d.

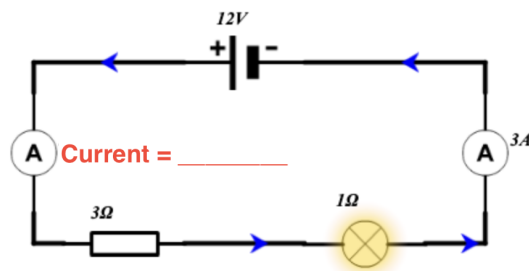


V = _____

· Parallel circuit. Each branch has the same voltage as the supply, so the unknown voltage = supply voltage.

2. Find the unknown current values in each circuit:

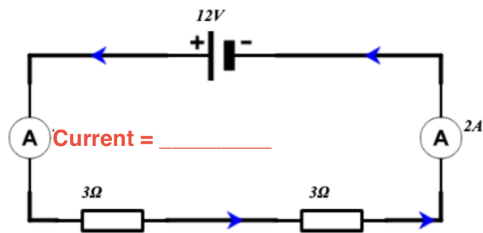
a.



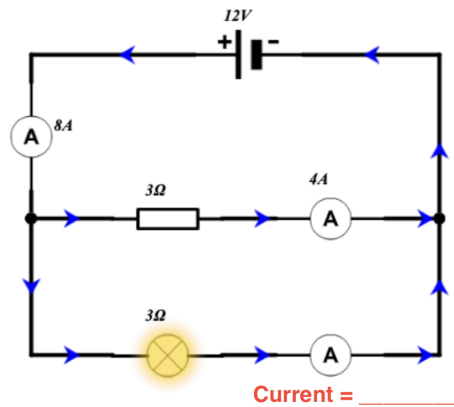
Current = _____

· Series circuit: the current is the same everywhere, so the unknown current = the known current.

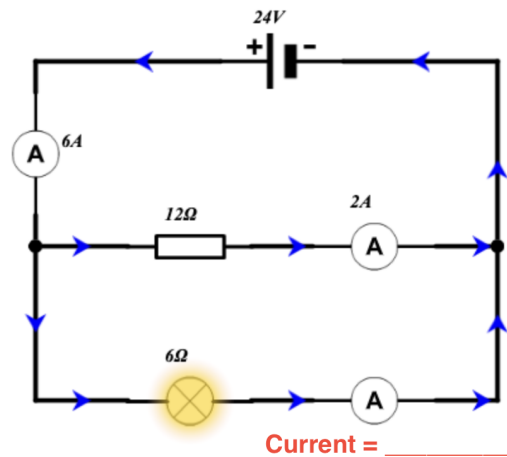
b.



c.

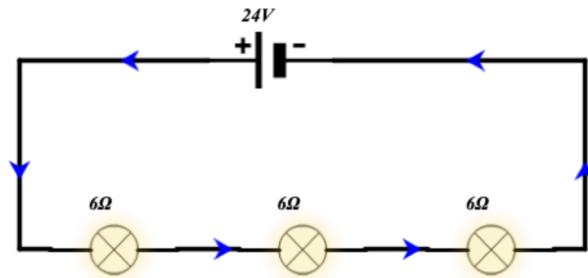


d.



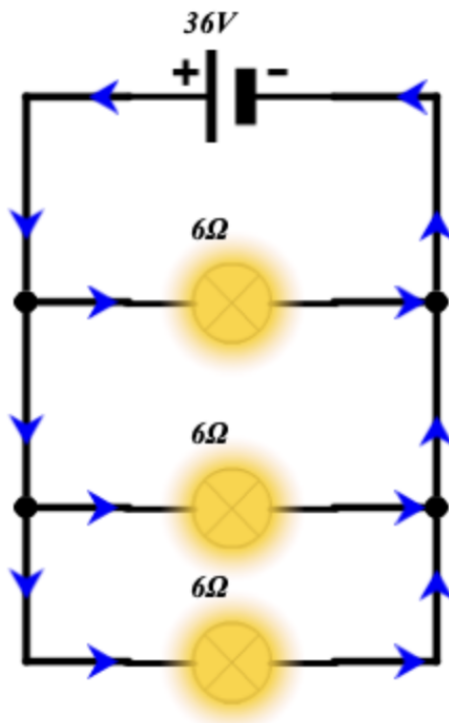
- Parallel circuit: the main current = sum of the branch currents, so subtract the known branch current from the main current to find the unknown one.
- Parallel circuit at a junction: the unknown branch current = main current - (sum of the other branch currents).
- Series circuit: the current is the same at every point, so the unknown current equals the current given elsewhere in the circuit.

3. What happens in this circuit if a bulb breaks? The circuit is broken, so no current flows and all the bulbs go out.



4. What happens in this circuit if a bulb breaks?

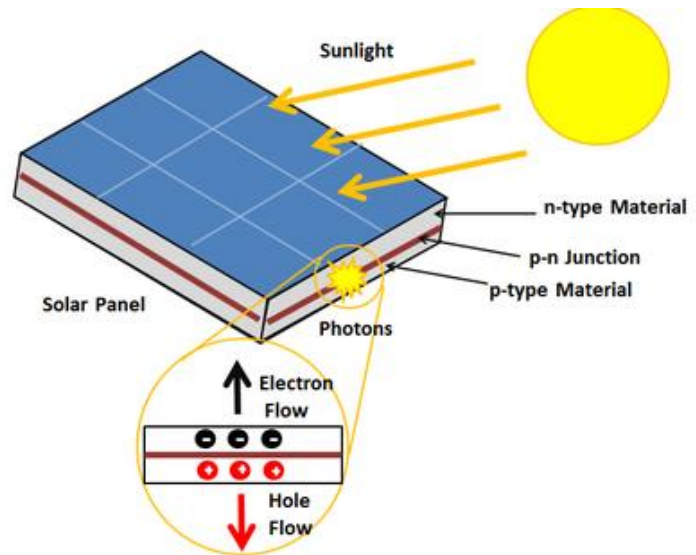
The other bulbs stay lit, because the current can still flow through the complete circuit of the other branches.



8P 6a Solar Cell Investigation

Introduction:

A solar cell is a device that converts **light energy** into **electrical energy**.
The voltage across the solar cell can be measured by a voltmeter.



Research Question:

How does changing the distance between a light source and solar cell affect the output voltage across the solar cell?

Control Variables:

Identify all appropriate control variables and write them down in the table below. Describe how they will be controlled in the experiment.

Control Variable	How will it be controlled /measured?
e.g. area of the solar cell	By using the same solar cell of a fixed area. It can be measured by using a ruler to measure the length and breadth and calculating using: $\text{area} = \text{length} \times \text{breadth}$.
Same light source	Use the same lamp and bulb, on the same power pack setting, for every reading.
Same solar cell	Use the same solar cell, kept perpendicular to the lamp and not moved between readings.
Same voltmeter	Use the same voltmeter to measure the output voltage each time.

Hypothesis:

Predict what will happen to the output voltage across the solar cell as you change distance between the solar cell and the light source? **Explain your prediction** by referring to scientific concepts and ideas. Include a diagram if that helps to aid your explanation.

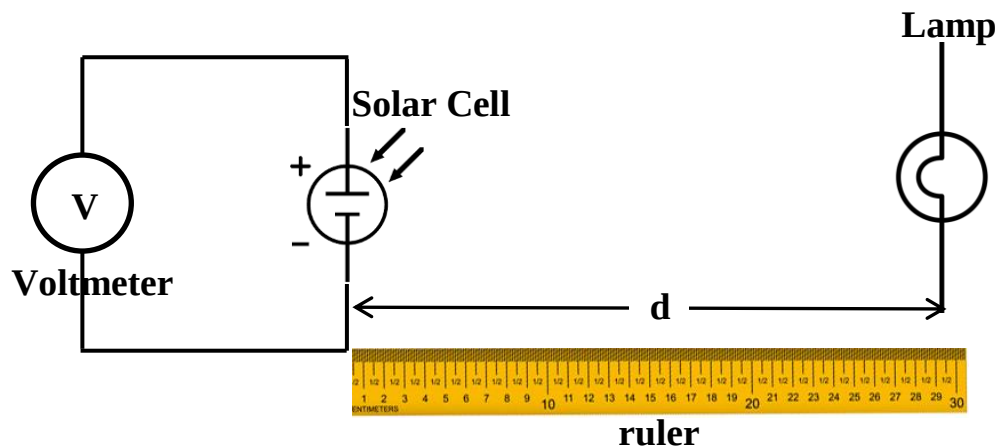
As the distance between the light source and the solar cell increases, the output voltage will decrease.

Because the light spreads out and becomes less intense as it travels,

so less light energy falls on the solar cell each second and a smaller voltage is produced.

Apparatus:

Solar cell, clamp, voltmeter, 6 electrical wires, power pack, bulb, meter stick

**Method:**

1. Set up the apparatus as above.
2. Decide on a suitable distance values to use, and add them to the table below. Ensure the range and increments are sufficient. You may do a quick trial to check this.
3. Use the clamp to ensure that the solar cell is perpendicular to the surface.
4. Move the lamp to the first specified distance by placing it at the appropriate place along the meter stick. Ensure that the distance is measure accurately.
5. Turn on the bench lamp
6. Measure the output voltage across the solar cell from the voltmeter and record the data in the table below.
7. Turn off the bench lamp.
8. Move the bench lamp to the next distance and repeat steps 3-6.
9. Repeat the experiment 2 more times, and write the results in the table. Calculate the mean (average) output voltage across the solar cell.

Safety:

Describe the safely precautions that you should take during the experiment:

1. Keep all water and liquids away from the electrical equipment.
2. Do not touch the lamp or bulb - it gets very hot.
3. Do not look directly at the lamp, as the light is very bright.
4. Keep the wires tidy so nobody trips over them.
5. Switch off the power pack when you are not taking a reading.

Results:

Distance (cm)	Output voltage across solar cell (V)			Average Voltage (V)
	Voltage 1	Voltage 2	Voltage 3	
10	6.4	6.3	6.5	6.4
20	4.2	4.1	4.3	4.2
30	2.9	2.8	3.0	2.9
40	2.1	2.0	2.2	2.1
50	1.5	1.6	1.4	1.5
60	1.1	1.0	1.2	1.1
70	0.8	0.9	0.7	0.8
80	0.6	0.5	0.7	0.6

Analysing Your Results:

Find the mean (average) by adding up all three results and then divide your answer by 3.
Show your results in a **line graph** and ensure that you address the following criteria:

- *Title of graph included*
- *Label on x and y axis with correct units*
- *Scales should be appropriate*
- *Points plots accurately and connect using curve/straight line of best fit*

Conclusions

What was the correlation between the distance between a light source and solar cell and the output voltage across the solar cell?

There is a negative correlation: as the distance between the lamp and the solar cell increases, the output voltage decreases.

The relationship is not linear - the voltage falls most quickly at the smaller distances.

Was your original hypothesis correct? If not, try to explain why and modify your explanation.

Yes, the hypothesis was correct - the output voltage decreased as the distance increased.

The results matched the prediction because light intensity decreases with distance,

so less light energy per second reaches the solar cell.

Evaluation

A few points to consider:

- Was this a fair test? (refer back to the section on control variables)
- Was it reliable?
- Were there any problems carrying out the experiment?
- How could this experiment be improved?

Yes, the test was fair because all the control variables were kept the same.

The same solar cell, lamp and voltmeter were used and the cell was kept perpendicular to the lamp.

The results were reliable because each distance was repeated three times.

It was reliable because each distance was repeated three times and the mean voltage was calculated; the repeat readings were close together (consistent).

The three repeats were close to each other, so the mean is trustworthy.

A problem was that the lamp got hot, which changed its brightness during the experiment.

Another problem was measuring the distance accurately from the lamp filament.

Light coming in from other sources in the room may have affected the readings.

Improvement: carry out the experiment in a dark room or shield it from other light.

Improvement: take more repeats and use the mean to reduce random error.

Improvement: fix a metre rule to the bench so the distance is measured accurately.

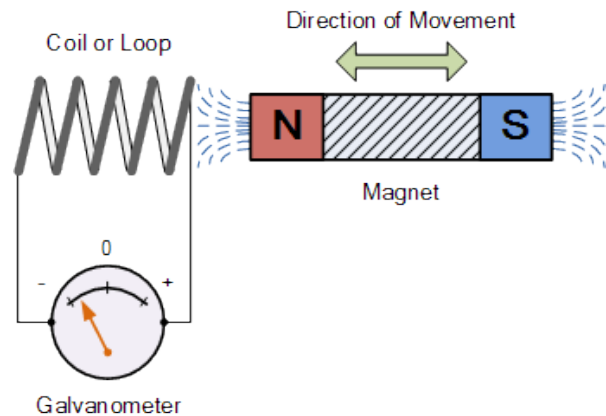
8P 7a Making Electricity

Introduction:

Making electricity is easy! Only 3 things are required:

1. A magnet
2. A coil of wire
3. Movement

When we make electricity this way, we say that we have induced a voltage in the coil.

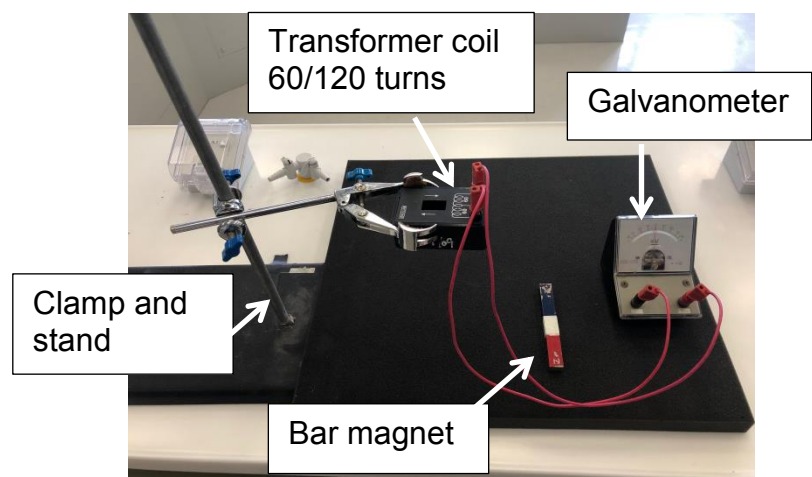


Aim:

To investigate the factors which affect the size of the induced voltage.

Apparatus:

- Transformer coil 60/120 turns
- Clamp
- Stand
- 2 wires
- Bar magnet
- Analogue milli-voltmeter
- Soft padding



Method:

1. Set up the apparatus as above.
2. Ensure that you are using 60 turns.
3. Place the magnet in the coil of wire and hold it stationary. What do you notice about the reading on the millivolt meter? Record your observation in the table below.
4. Place the north pole of the magnet above the coil of wire and drop it through the coil. Observe the direction the voltmeter needle moves. Repeat this, but position the south pole of the magnet above the coil. Compare the direction of the needle and record this in the table below.
5. Place the north pole of the magnet above the coil of wire. Drop it and observe the voltage. Repeat this, but start with the magnet further in the coil of the wire. This means the wire is moving slower past the coil. Observe the induced voltage and compare with the voltage when the magnet was moving more quickly. Record your results in the table below.
6. Change the number of coils to 120 turns. Drop the North Pole through first from the same height. Compare the size of the induced voltage between 60 and 120 turns.

Observations:

Change	Observation
Hold the magnet stationary in the coil.	The millivoltmeter reads zero - no voltage is induced because there is no movement.
North pole or south pole first.	The needle deflects in opposite directions - reversing the pole of the magnet reverses the induced voltage.
Move the magnet quickly then slowly through the coil.	The induced voltage is <u>greater</u> (smaller) if the coil is moving faster.
60 turns or 120 turns at the same speed.	The induced voltage is <u>greater</u> (smaller) if more turns are used.

Conclusions:

List the ways that you can increase the size of the induced voltage.

Move the magnet faster through the coil.

Use a stronger (more powerful) magnet.

Use more turns on the coil (120 turns instead of 60).

Move the magnet so it cuts through more magnetic field lines each second.

Extension:

As the magnet passes through the coil, you should observe the millivolt meter needle move in one direction before swinging back in the opposite direction. In other words, the induced voltage values are positive then become negative.

Using your knowledge of electromagnetic induction, explain what causes this to happen?

The induced voltage changes direction as the magnet passes through the coil.

As the magnet enters, the field lines cut the coil one way, giving a positive voltage.

As it leaves the other side, the lines cut the coil in the opposite direction,

so the induced voltage becomes negative - the voltage (and current) is alternating.

8P 7b Dynamos and Generators

Dynamos

Your teacher will demonstrate a dynamo in action.

Describe how it works in the space below:

A dynamo has a coil of wire that spins between the poles of a magnet.

The movement of the coil in the magnetic field induces a voltage,

so an electric current flows in the coil and around the circuit.

The faster the coil spins, the bigger the induced voltage.



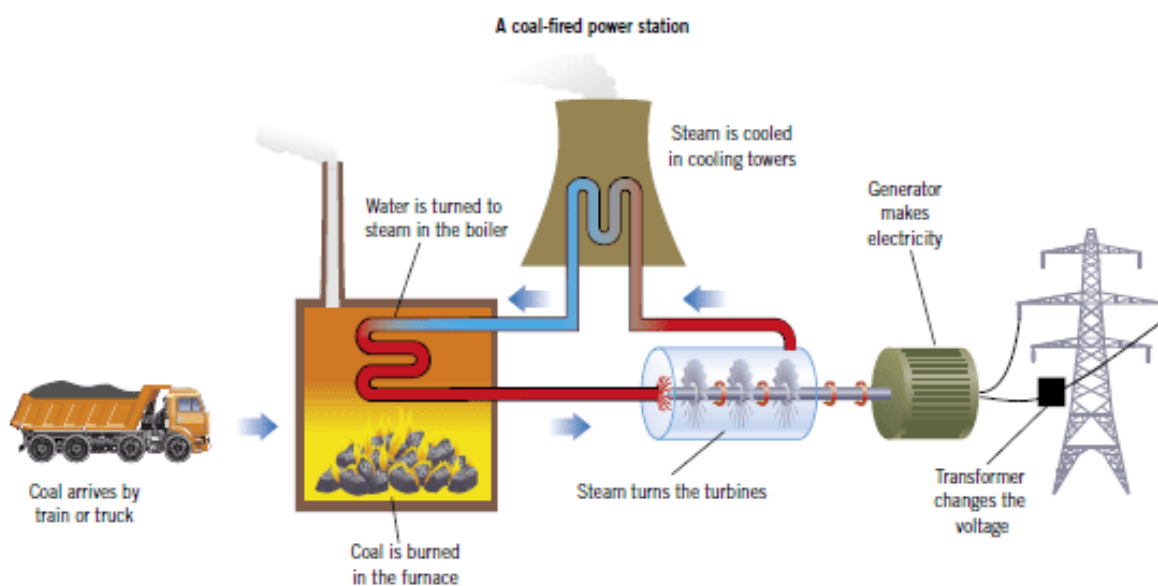
Generators and Power Stations

We can take the principles of making electricity and apply them to make electricity on a larger scale in power stations. It is important to remember that the same three ingredients are required:

- A magnet
- A coil of wire
- Motion

Here is a diagram of coal-fired power station.

Your teacher will explain how it works.



Here is a link that also explains:

<https://www.bbc.com/bitesize/guides/zbsdmp3/revision/1>

Write a description of how it works below. Try to include the energy transfers that happen. Using bullet points will be helpful:

Coal is burned in the boiler - chemical energy is transferred to heat energy.

The heat boils water to make high-pressure steam.

The steam pushes the blades of the turbine, so heat energy becomes kinetic energy.

The turbine spins a generator, which changes kinetic energy into electrical energy.

The generator works because a coil spins inside a magnetic field, inducing a voltage.

The steam is cooled back to water by the cooler (condenser) and used again.

A transformer steps up the voltage so the electricity can be sent along the grid.

Energy is wasted as heat and sound in the turbine, generator and cables.

So the overall efficiency of the power station is less than 100%.

Extension:

The voltage induced in dynamos and generators is alternating current (a.c.). Find out what this means and why is produced in generators and dynamos.

a.c. means alternating current - the current keeps changing direction.

The coil spins, so it first cuts the magnetic field lines one way and then the other,

which reverses the induced voltage and current many times each second.

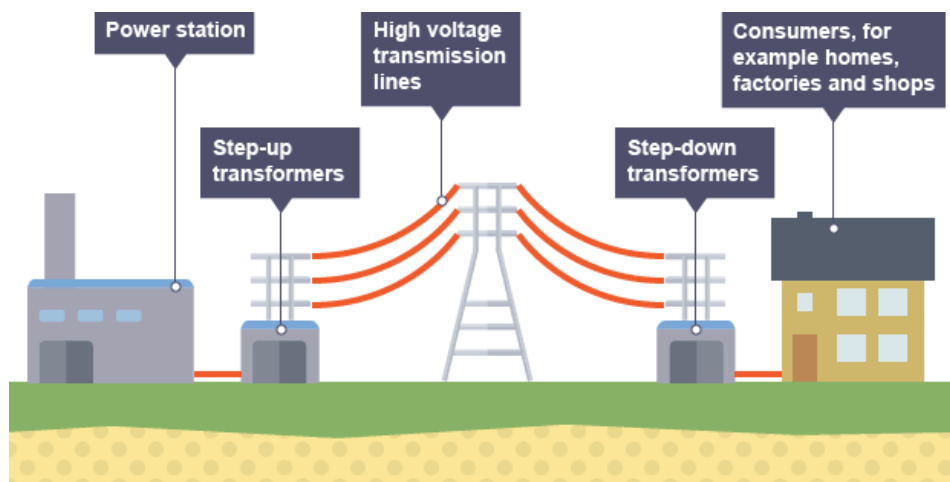
8P 7c Power Transmission

Introduction:

Once electricity has been generated in a power station, it needs to be transmitted from the power station or generator to our homes. This is done using a system of pylons and cables and transformers. In HK a lot of our transmission cables are underground as well.

Model Transmission Line:

Your teacher will show you a model transmission line. It should look similar to the diagram below:



How it Works

Some energy is 'lost' in the transmission lines as heat, due to resistance in the cables and a small amount of sound energy. If you have ever stood near a pylon you may have heard a faint buzzing noise, as some energy is transferred to sound.

We can reduce the loss of energy as heat by ensuring that the electricity in the transmission lines has a lower current. When there is a low enough current, less electrical energy is lost as heat. This is done by using a step-up transformer to reduce the current and therefore the energy lost as heat. The step-up transformer does however increase the voltage to very high voltages (up to 500,000V) which makes it very dangerous. Before the voltage can be safely used by consumers, a step-down transformer is used to reduce the voltage to make it safe.

Questions:

Having seen the model power line and read the description above, answer the following questions in full sentences:

1. How does electricity get transported from power stations to homes?

Electricity is carried from power stations to homes by a network of pylons, _____
cables (mostly underground in Hong Kong) and transformers. _____

2. How is energy 'lost' in the transmission lines?

Energy is lost as heat because the transmission cables have resistance.

so as the current flows through them some electrical energy is transferred to heat.

A small amount of energy is also transferred to sound - cables buzz near pylons.

This energy is wasted because it never reaches the consumer.

3. Describe how the energy loss can be reduced.

Reduce the current flowing in the transmission lines.

because a lower current means less electrical energy is lost as heat.

This is done with a step-up transformer, which increases the voltage

and so lowers the current for the same power transmitted.

4. Describe the role of the step-up transformer.

The step-up transformer increases the voltage to very high values (up to 500 000 V).

This reduces the current, so less energy is wasted as heat in the cables.

It is placed between the power station and the transmission cables.

5. Why must a step-down transformer be used before?

Because the transmission voltage is dangerously high (up to 500 000 V).

the step-down transformer reduces it to a safe voltage (about 220 V) for consumers.

8P 8a Energy – Identifying Fossil Fuels



Hydro (water) - electric



Wind turbines



Coal



Oil



Solar



Gas

Which picture above is an example of a fossil fuel? Circle your answers.

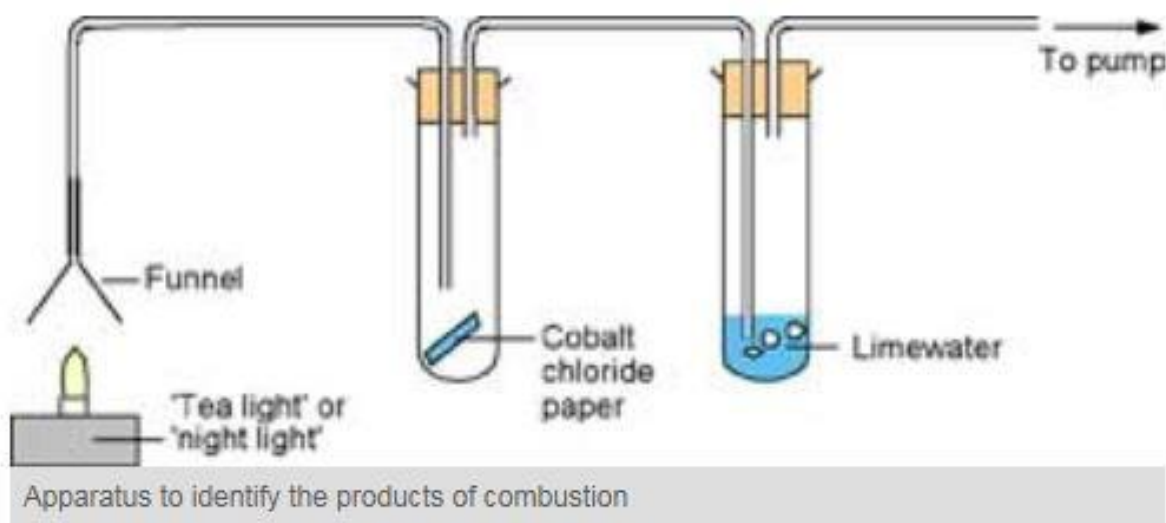
Explain why those that you have circled are fossil fuels.

They were formed from the remains of dead plants and animals buried underground over millions of years, and they release carbon dioxide when they are burned.

8P 8b Production of carbon Dioxide

Your teacher will demonstrate the following experiment that shows how fossil fuels release carbon dioxide gas when burned.

Website Resource: <http://www.rsc.org/learn-chemistry/resource/res00000707/identifying-the-products-of-combustion?cmpid=CMP00005148>

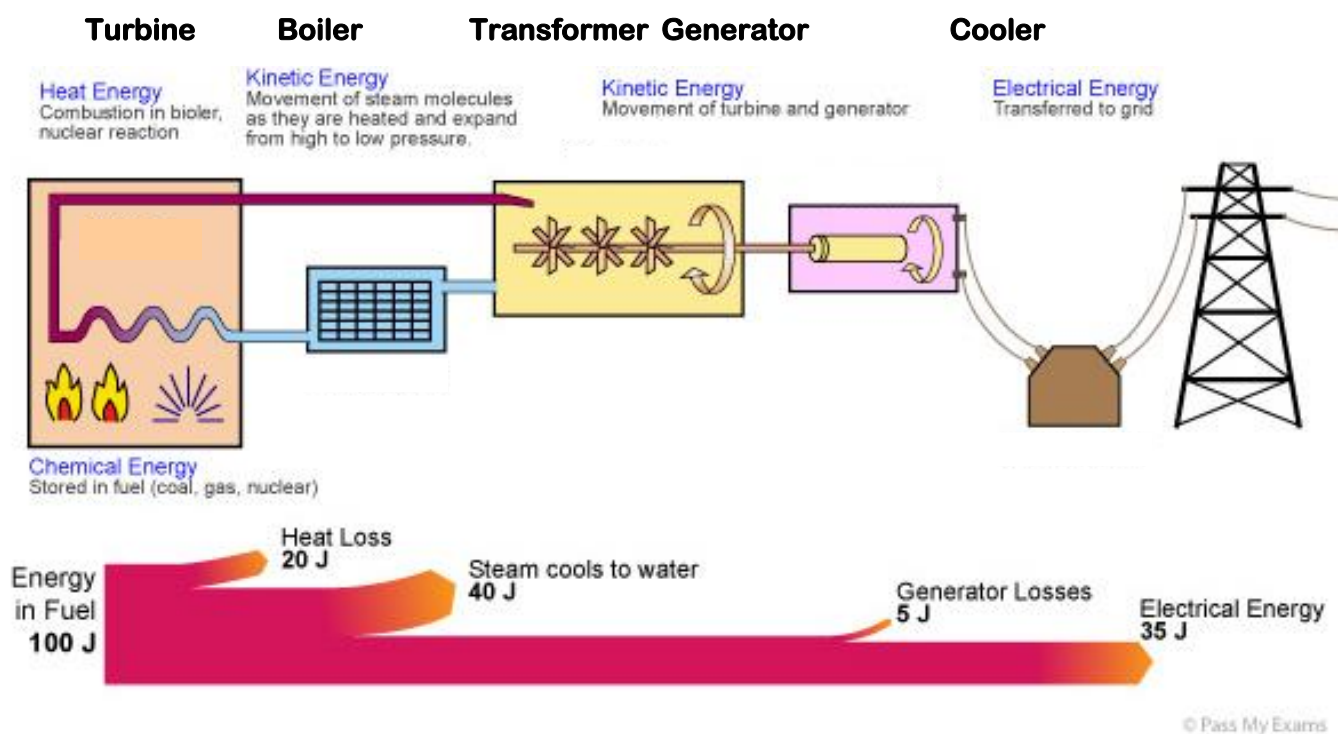


Describe your observations as the fuel is burned. State what fuel is being used.

Fuel used: a hydrocarbon fuel such as ethanol or paraffin. It burns with a yellow flame, and the limewater turns milky (cloudy), showing that carbon dioxide gas is produced.

8P 8c Power Stations

1. Add the following labels to this powerstation diagram. Write them next to the part of the power station that does this.



2. Calculate the efficiency of the power station in turning Energy in Fuel to Electrical Energy.

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100\%$$
For example, if 100 J of energy in the fuel gives 35 J of electrical energy, $\text{efficiency} = \frac{35}{100} \times 100 = 35\%$.

3. The table below has information about 5 different types of power station. Draw lines to link up the information about each type of power station.

<u>Type of Power Station</u>	<u>Advantage</u>	<u>Disadvantage</u>
Coal, Oil and Gas	Produces no CO ₂ , even in construction.	Very expensive to construct, and potentially very dangerous.
Nuclear	Energy source is totally free and available daily.	Damages very large areas of beautiful countryside.
Wind	Power stations can also provide leisure facilities.	Sometimes considered ugly and not available every day.
Solar	Quick to construct and operates all the time.	Not available 24 hours a day.
Hydroelectric Dam	Very small quantities of waste are made, and no CO ₂ .	Produces large amounts of CO ₂ which increases global warming.

Coal, Oil and Gas - Advantage: Quick to construct and operates all the time. Disadvantage: Produces large amounts of CO₂ which increases global warming.

Nuclear - Advantage: Very small quantities of waste are made, and no CO₂. Disadvantage: Very expensive to construct, and potentially very dangerous.

Extension (optional)

Write a letter to the government explaining why we should use renewable resources, rather than coal, gas or oil to generate electricity.

Wind - Advantage: Produces no CO₂, even in construction. Disadvantage: Sometimes considered ugly and not available every day.

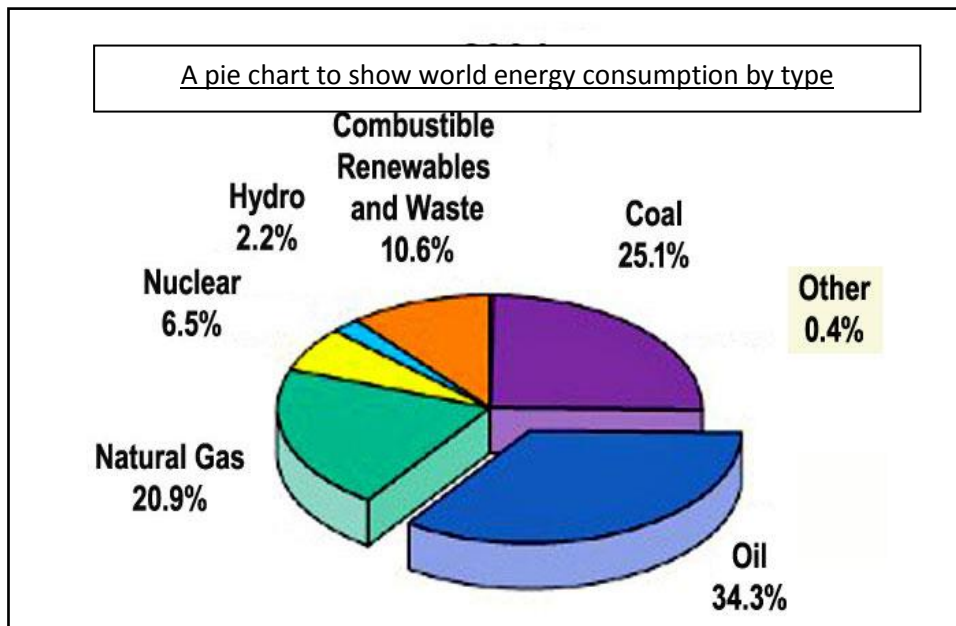
Solar - Advantage: Energy source is totally free and available daily. Disadvantage: Not available 24 hours a day.

Hydroelectric Dam - Advantage: Power stations can also provide leisure facilities. Disadvantage: Damages very large areas of beautiful countryside.

Burning coal, gas and oil releases carbon dioxide, which increases global warming and climate change. Renewable sources such as wind, solar and hydro produce no CO₂ and will never run out.

8P 8d Problems with Fossil Fuels

A fossil fuel is A fossil fuel is a fuel formed from the remains of dead plants and animals
buried underground for millions of years, such as coal, oil and natural gas
They are non-renewable, because they will eventually run out



The pie chart shows that The pie chart shows that fossil fuels (coal, oil and gas) supply most of the world's energy.
Oil is the largest single source, followed by coal and gas.
Renewable sources such as hydro, nuclear, wind and solar provide only a small share
So the world still depends heavily on fossil fuels.

There are several reasons for the trend such as
They are cheap and easy to extract.
They release a lot of energy when burned and are easy to store and transport
Power stations that use them are quick to build and can run all day and night
Renewable sources such as wind and solar are not available all the time
and can be expensive to build
The world's demand for energy has also increased rapidly

The pie chart shows that the world needs to use alternative sources of energy because
It produces a greenhouse gas that causes global warming and climate change
Fossil fuels are also non-renewable and will eventually run out
Alternative (renewable) sources are cleaner and will not run out

8P 9a Consequences of Global Warming

Consider the following:

Oceans absorb more heat than the land, and water expands when it is heated so this could cause sea levels to rise.

Glaciers (big 'rivers' of ice) and sea ice could also melt leading to sea level rise.

Coastal cities could flood destroying valuable land.

The climate of places could change. Places which usually get lots of rain may get hotter and drier.

Beaches could disappear because of sea level rises.

Many animals would be threatened with extinction (dying out) because of the heat and the loss of their habitats (homes).

There would be a greater risk of forest fires due to the increase in temperature.

There would be more droughts (Lack of rainfall) making it hard to grow crops in particular places.

There could be an increase in the number of storms and extreme weather events because of the changes in heat.

Your task:

- Divide your file worksheet into 2 – one column for **human impacts** and one for **environmental impacts**.
- Take each point above in turn and decide if it would affect humans, the environment or both.
- Summarise the point in your chosen column as to why the impact would occur.
- You may work in groups of two or three to complete this task.

Human impacts: coastal cities flood, land and beaches lost, crops fail in droughts, storm damage. Environmental impacts: ice melts, sea level rises, habitats lost so animals die out, forest fires, droughts.

Example: sea levels rise because the oceans absorb heat (water expands when heated) and glaciers/sea ice melt - a cause then its effect.

The Greenhouse Effect

Some solar radiation is reflected by the Earth and the atmosphere.

Some of the infrared radiation passes through the atmosphere. Some is absorbed and re-emitted in all directions by greenhouse gas molecules. The effect of this is to warm the Earth's surface and the lower atmosphere.

Most radiation is absorbed by the Earth's surface and warms it.

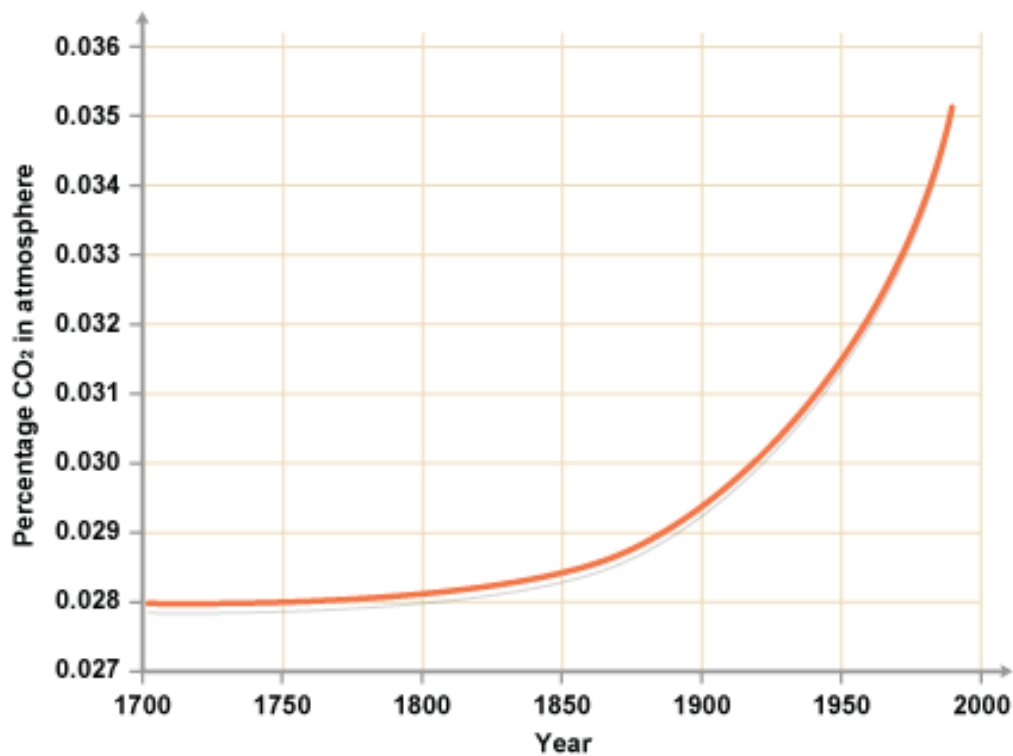
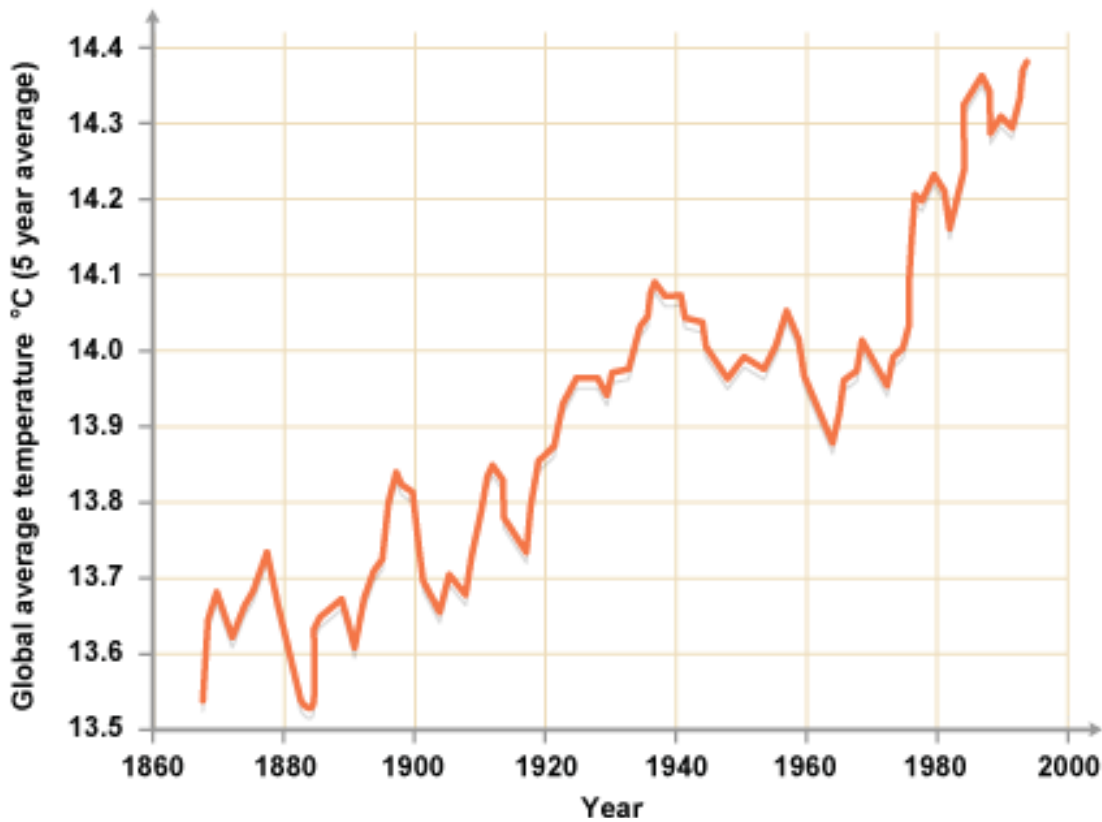
Atmosphere

Earth's surface

Infrared radiation is emitted by the Earth's surface.

8P 9b Global Temperature and Carbon Dioxide

Examine the graphical data below lined to Global Warming.



· Graph 1: the global average temperature rises steadily over the years shown. Graph 2: atmospheric carbon dioxide also rises steadily - both graphs trend upwards together.

Using the graphs answer the questions below:

Explain:

- What trend does the first graph show?

The first graph shows an upward trend - the global average temperature has steadily increased over the years shown.

- What has been the overall increase in temperature?

An overall increase of about 0.8 °C (roughly 1 °C) between the start of the graph and the present day.

- What trend does the second graph show?

The second graph also shows an upward trend - the concentration of carbon dioxide in the atmosphere has steadily increased over the same years.

Extension

Do the two graphs have any relationship? How could they be connected? Discuss.

Yes. Both graphs rise at the same time so they are linked: as carbon dioxide (a greenhouse gas) increases it traps more heat in the atmosphere, so the temperature rises - the enhanced greenhouse effect.

8P 9c Renewable Energy Resources

Type of energy	Where it is from	Advantages	Disadvantages
Solar	Energy from sunlight is captured in solar panels and converted into electricity.	Potentially infinite energy supply. Single dwellings can have own electricity supply.	Manufacture and implementation of solar panels can be costly.
Wind	Wind turbines (modern windmills) turn wind energy into electricity.	Can be found singularly, but usually many together in wind farms. Potentially infinite energy supply.	Manufacture and implementation of wind farms can be costly. Some local people object to on-shore wind farms, arguing that it spoils the countryside.
Tidal	The movement of tides drives turbines. A tidal barrage (a kind of dam) is built across estuaries, forcing water through gaps. In future underwater turbines may be possible out at sea and without dams.	Ideal for an island such as the UK. Potential to generate a lot of energy. Tidal barrage can double as a bridge, and help prevent flooding.	Construction of barrage is very costly. Only a few estuaries are suitable. Opposed by some environmental groups as having a negative impact on wildlife. May reduce tidal flow and impede flow of sewage out to sea.
Wave	The movement of seawater in and out of a cavity on the shore compresses trapped air, driving a turbine.	Ideal for an island country. More likely to be small local operations, rather than done on a national scale.	Construction can be costly. May be opposed by local or environmental groups.
Geothermal	In volcanic regions it is possible to use the natural heat of the earth. Cold water is pumped under ground and comes out as steam. Steam can be used for heating or to power turbines creating electricity.	Potentially infinite energy supply. Used successfully in some countries, such as New Zealand and Iceland.	Can be expensive to set up and only works in areas of volcanic activity. Geothermal and volcanic activity might calm down, leaving power stations redundant. Dangerous elements found underground must be disposed of carefully.

Type of energy	Where it is from	Advantages	Disadvantages
Hydrological or Hydroelectric Power (HEP)	Energy harnessed from the movement of water through rivers, lakes and dams.	Creates water reserves as well as energy supplies.	Costly to build. Can cause the flooding of surrounding communities and landscapes. Dams have major ecological impacts on local hydrology.
Biomass	Decaying plant or animal waste. An organic material, which can be burned to provide energy, eg heat, or electricity. An example of biomass energy is oilseed rape (the fields of yellow flowers you see in the UK in summer), which produces oil. After treatment with chemicals it can be used as a fuel in diesel engines.	It is a cheap and readily available source of energy. If replaced, biomass can be a long-term, sustainable energy source.	When burned, it gives off atmospheric pollutants, including greenhouse gases. Biomass is only a renewable resource if crops are replanted.
Wood	Obtained from felling trees, burned to generate heat and light.	A cheap and readily available source of energy. If the trees are replaced, wood burning can be a long-term, sustainable energy source	When burned it gives off atmospheric pollutants, including greenhouse gases. If trees are not replanted then wood is a non-renewable resource.

Research Points – Renewable Energy Technology

Research your chosen technology: how it works, where it can be sited, cost, output, advantages over fossil fuels, disadvantages (visual impact, wildlife, power lines) and data sources to cite.



MYP 2: Year 8 Science

Criterion D Assessment - Reflecting the Impacts of Science

UNIT:	Energy Production and Climate Change
ASSESSMENT TITLE:	Renewable Energy Technology
Inquiry Statement	Exploring changes in the environment brought about by energy production and investigating the consequences on humankind .
Key Concept:	Change
Related Concepts:	Energy and Consequences

Assignment Instructions (GRASPS)

Goal	Produce a persuasive Powerpoint presentation or Video providing information on a renewable energy source and highlighting the pros and cons of this energy source compared with burning fossil fuels.
Role	You are part of an environmental group who wishes to persuade the government to introduce legislation for using a renewable source of energy to replace fossil fuels.
Audience	The Government and the local population of your country.
Situation	You are campaigning for renewable energy to replace fossil fuels where you live.
Product	Powerpoint or video presentation.
Success Criteria for Assessment	You will clearly explain the advantages of the renewable energy source compared with fossil fuels and you will have data to support your argument. Presentation will be easy to read, well-illustrated and contain clear and relevant data explained using good scientific vocabulary.

Other Information:

You MUST provide a Bibliography and cite all your sources.

Due Date: this will be set by your teacher.

Level Descriptors and Task Specific Clarifications for Criterion D:

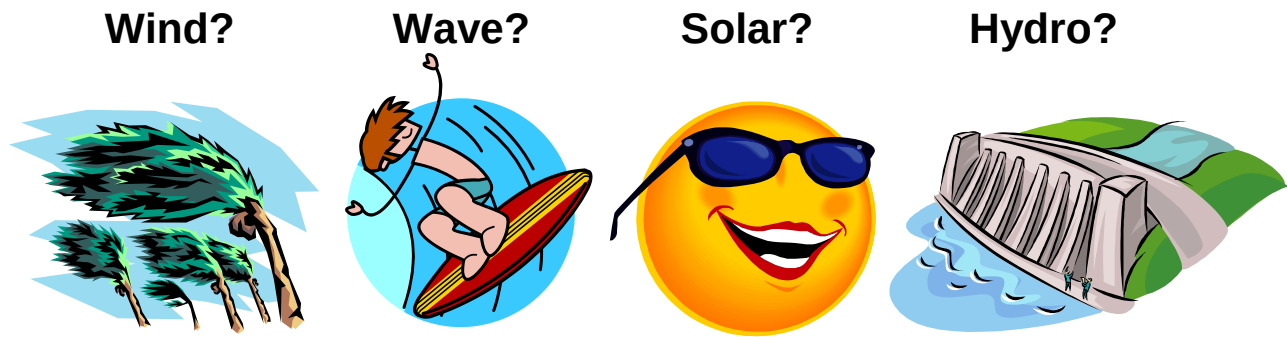
Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	No data or clear evidence provided.
1-2	The student is able to, with limited success : i. state the ways in which science is used to address a specific problem or issue ii. state the implications of using science to solve a specific problem or issue, interacting with a factor iii. apply scientific language to communicate understanding iv. document sources.	ii. Student simply states the problem of using fossil fuels ii. Simply state that the renewable energy source is better. No explanation or relevant data given. iii. Little evidence of scientific vocabulary being used. iv. Source of information given, but not correctly referenced.
3-4	The student is able to: i. Outline the ways in which science is used to address a specific problem or issue ii. Outline the implications of using science to solve a specific problem or issue, interacting with a factor iii. sometimes apply scientific language to communicate understanding iv. sometimes document sources correctly.	i. At least one problem associated with burning fossil fuels is outlined. Some data is given, but source may not be referenced. ii. Student provides at least 1 reason to explain why the renewable source is an improvement. iii. Some scientific vocabulary is used. iv. Bibliography has only one source referenced, but this may not be accurately formatted.
5-6	The student is able to: i. Summarize the ways in which science is used to address a specific problem or issue ii. Describe the implications of using science to solve a specific problem or issue, interacting with a factor iii. usually apply scientific language to communicate understanding clearly and precisely iv. usually document sources correctly.	i. Problems associated with burning fossil fuels are summarized, but not explained fully. Some data that supports the argument is given. ii. The advantages of the renewable source are described, but not fully explained. There is some mention of energy types. iii. Some good scientific vocabulary is used. iv. At least 3 sources are referenced in APA format
7-8	The student is able to: i. Describe the ways in which science is applied and used to address a specific problem or issue ii. Describe and summarize the implications of using science and its application to solve a specific problem or issue, interacting with a factor iii. consistently apply scientific language to communicate understanding clearly and precisely iv. document sources completely	i. The student presents a clear explanation of the problems associated with fossil fuels. Equations are given to explain the problems burning fossil fuels cause. Include 1 piece of verifiable data to support the argument for changing from fossil fuels to the renewable source. ii. Contrast these with the advantages of the renewable source, describing and summarising how this works (eg energy transfers) iii. Explanations involve excellent scientific vocabulary. iv. Correct bibliography is provided citing at least 5 reputable and relevant sources, in APA format.

8P10a Renewable Energy Sources - Research Template

Questions	Notes
What are the problems associated with burning fossil fuel? Burning fossil fuels releases carbon dioxide, a greenhouse gas that causes global warming and climate change; it also releases sulphur dioxide (acid rain) and particulates, and the fuels will run out. 1. Describe in detail the problems associated with burning fossil fuels. 2. What evidence is there to substantiate the problems you have described? Evidence to support the problems described includes rising atmospheric CO2 measurements, global temperature records, melting polar ice and sea-level rise reported by scientists. Evidence: measured rise in atmospheric CO2 (Keeling curve), global temperature records, melting glaciers and ice caps, rising sea levels, IPCC reports and satellite data.	Burning fossil fuels releases CO2 (a greenhouse gas) which traps heat, causing global warming, plus acid rain and smog. Evidence: CO2 up from 280 to over 410 ppm since 1850 while temperature rose about 1 °C.
Renewable Solution 3. Choose a renewable energy technology. 4. Describe in detail how it works. Be sure to describe the energy transfers taking place. Example (wind): moving air turns the rotor blades, the rotor drives a generator, transferring kinetic energy to electrical energy, which is sent to the National Grid.	e.g. wind power: moving air (kinetic energy) turns the turbine blades, which spin a generator, transferring kinetic energy to electrical energy. Cables and the National Grid carry it to homes. Choose one, e.g. an offshore wind farm (others: solar farm, hydroelectric, tidal, geothermal, biomass).
Impact 5. What are the benefits of using this renewable technology compared to burning fossil fuels? 6. Are there any other advantages to society or the environment? 7. What are the disadvantages of using this renewable solution? Hint: Think about the impact that it has on the environment, wildlife the economy and society in general. Advantages for society include new jobs and cheaper bills; for the environment, less pollution and no greenhouse gases released when generating. Other advantages: local jobs and industry, energy security (no imported fuel), and less air pollution so better public health. Disadvantages: return is intermittent (no wind or sun = no electricity)	Benefits: no greenhouse gases, free fuel that will not run out, low running costs. Disadvantages: high build cost, visual impact, noise, effects on wildlife, needs a windy site. Benefits: no carbon dioxide or polluting gases are released when generating, the energy source is renewable and free, and running costs are low. Compared with fossil fuels it produces no CO2 or acid rain gases, the fuel costs nothing and will not run out, so it is more sustainable. Energy flow: kinetic (wind) -> kinetic (turning turbine and generator) -> electrical, with heat and sound wasted. Sketch the turbine, generator and transmission cables. high initial cost, needs land or sea space, and can affect wildlife and views.
Additional notes/diagrams Disadvantages of the renewable solution: unreliable/intermittent supply needing backup or storage, high building cost, and visual/noise or habitat impacts.	

References

8P 11a Renewable Energy Solutions



YOU DECIDE!

Task:

Communities are concerned about climate change and future energy supplies. They have decided to develop renewable energy for your town.

Your task is to choose suitable forms of renewable energy for your area. You should try to provide the greatest possible number of homes with 'clean' electricity, at the lowest possible cost.

However - it is not quite that simple! Different types of renewable energy have their advantages and disadvantages.

They may also have strict requirements to make them work. You should carefully study the 'Power Facts' and your map to decide from the PowerPoint you teacher has shown you which sites would be suitable for the different types of energy.

Other points you need to consider include:

- **Visual impact** – will it be a 'blot on the landscape'?
- **Natural heritage** – how might it affect plants and animals?
- **Human activity** – will it cause difficulties for people? Who will you need to consult?
- **Archaeology** – are there important historic sites nearby?
- **Road access** – is your proposed site accessible?
- **Electricity transport** – if your site is remote, you will need more power lines to take the electricity to peoples' homes

Remember – there are no right or wrong answers. Discuss the issues carefully among your group – you may need to make compromises!

To start, it is sensible to consider as many feasible solutions as possible before you choose. Combinations of different types of renewable energy solutions are permitted.

Human activity · noise, traffic and land use may upset local people; consult local residents, the council/planners, landowners and environmental agencies.

You need to justify your choice and explain why you haven't selected other options.

Archaeology · check for listed buildings, ancient monuments or buried archaeology nearby, then avoid, protect or record the site.

Use the space below to make notes.

Notes: for each energy type list site requirements, output (homes supplied), cost, visual, natural-heritage, archaeology and access issues, and whether new power lines are needed; then rank and combine the best options.

8P 11b Renewable Energy Solutions

Presentation:

Your group now needs to present your renewable energy solution.

The presentation should contain the following:

1. Your location map with the renewable energy solution(s) drawn on it.
2. The total cost of your solution. Are there any additional cost such as roads and electricity transmission lines?
3. A justification of why you have chosen that particular solution. Why is it best?
4. An explanation as to why you didn't choose other possible solutions.

The total cost includes the turbines or panels plus extra costs such as new access roads and electricity transmission lines.



For points 3 and 4 above, you should also discuss the following:

- **Visual impact** – will it be a ‘blot on the landscape’?
- **Natural heritage** – how might it affect plants and animals?
- **Human activity** – will it cause difficulties for people? Who will you need to consult?
- **Archaeology** – are there important historic sites nearby?
- **Road access** – is your proposed site accessible?
- **Electricity transport** – if your site is remote, you will need more power lines to take the electricity to peoples’ homes

Justify the choice by comparing cost, output, reliability, environmental impact and location · e.g. wind is cheap per kWh but intermittent.

Justify: 'We chose offshore wind and solar because our coast is exposed and windy and the site is sunny, giving the most clean electricity for the lowest cost. We rejected a tidal barrage because it is very costly and would harm the estuary wildlife.'

Natural heritage · turbine blades can kill birds and bats and disturb habitats; careful siting and surveys reduce the damage.

Human activity · noise, traffic and access may upset local people; consult residents, the council/planners and landowners before building.

Archaeology · check for listed buildings, ancient monuments or buried remains nearby and avoid or protect the site.

Road access · heavy equipment needs suitable roads; a remote site may need new roads built, increasing the cost.

8P 12a Presentation Skills Self Assessment Rubric

Complete the following self-evaluation by ticking the boxes.

Reflect on your/your partner's performance on this task and try to set some targets to improve.

N = Novice **R= Reasonable** **G = Good** **E = Expert**

	N	R	G	E
1. Structure: Your project or presentation is really well organized and structured. This helps you communicate your understanding to others.				
2. Knowledge and understanding: You really understand the concepts that you are presenting or discussing.				
3. Application of Science: You apply what you have learnt to explain the uses of science to solve a problem or an issue.				
4. Implications of Science: You consider the positive and negative effects of Science on this problem or issue.				
5. Use of Scientific Words & Terminology: You use lots of important and relevant scientific terms & words. This helps others understand your work.				
6. Use of Scientific Diagrams, Photo's, Video clips, and Simulations: You use relevant digital media to help explain concepts and ideas.				
7. Clarity of Communication: Your written, oral or digital media presentation is really clear and easy for others to follow. For oral presentations, your tone, speed and volume of voice is confident and appropriate for your audience.				
8. Engaging and Interesting: You make your presentation interesting and show lots of enthusiasm and passion. You use words, phrases, pictures or media applications that really capture the imagination of your audience.				
9. Reading and use of Sources: You have done lots of background reading to help you make your presentation. You quote at least 2 or 3 sources of information at the end of your presentation.				
10. Summarizing: You are able to sum up concisely, so that your audience is clear what your final thoughts, opinions or conclusions are.				
11. Quality of final product: Your presentation or project is of a really high standard. Your audience, either reading or listening, seems to really understand what you are trying to communicate.				

Highlight a few of these skills as targets for your future development. If you want, give yourself a grade and discuss these targets with your teacher/tutor/parent or a friend.

Mainly Expert = (8)

Mainly Expert/Good = (7)

Mainly Good = (6)

Mainly Good/Reasonable = (5)

Mainly Reasonable = (4)

Mainly Novice/Reasonable = (3)

Mainly Novice = (1 or 2)

Grade =

(if appropriate)

Student/peer Targets:

The most important thing I learnt was ~~that~~ renewable energy resources will not run out and give off no greenhouse gases, so they can replace fossil fuels.

This could be better if I ~~used more scientific vocabulary~~ and included data with references to support my argument.

8P 13a Summary Notes

Energies in action

- heat energy
- light energy
- sound energy
- electrical energy
- kinetic (movement) energy.

Stored energy

Some energy has to be stored so that it is ready for use when we need it.

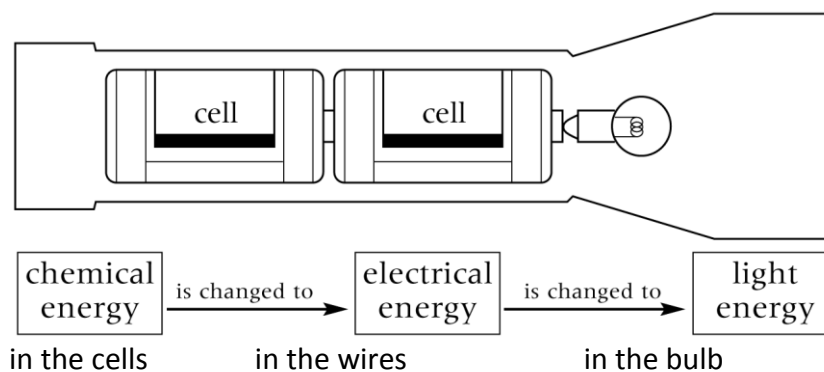
- Chemical energy is stored in food, fuels and cells.
- Gravitational potential energy is stored in high up things.
- Strain energy is stored in stretched or squashed things.
- Nuclear energy is stored inside atoms.

How is energy measured?

Energy is measured in **joules (J)** or **kilojoules (kJ)**. A kilojoule is 1000 joules.

Energy changes

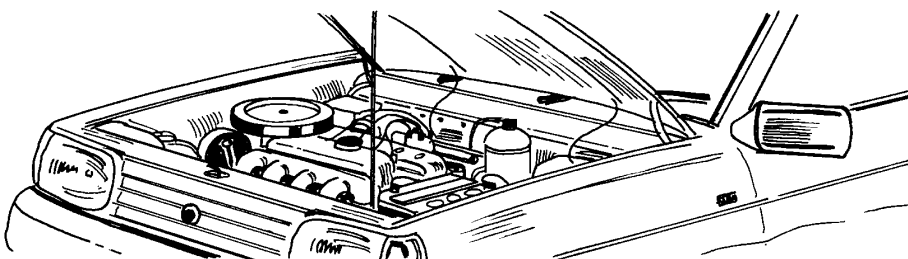
Energy needs to be changed to be useful.



An energy flow diagram.

Energy is measured in joules (J) or kilojoules (kJ); 1 kJ = 1000 J. It is calculated using $E = P \times t$ or measured with a joule meter.

Many energy changes take place in everyday life. Often wasted energy is produced in the forms of **heat** or **sound**.



A car engine produces kinetic energy, which is useful. It also produces heat and sound

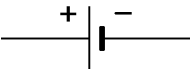
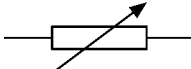
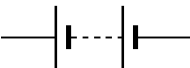
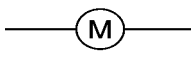
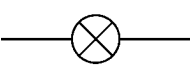
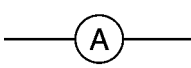
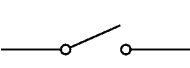
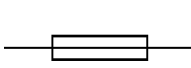
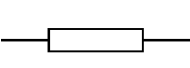
Energy cannot be made or destroyed, but can only be changed from one form to another. This is the **law of conservation of energy**.

Circuits

Electricity is a flow of **electrons**. Electricity can flow through **conductors** but not through **insulators**. **Metals** are good conductors of electricity.

A complete **circuit** is needed for electricity to flow.

We use symbols when we draw circuits:

Component	Symbol	Component	Symbol
cell		variable resistor	
battery of cells		motor	
bulb		ammeter	
switch		fuse	
resistor		Voltmeter	Draw here:

Current

The **current** is the amount of electricity flowing in the circuit. The units for current are **amps (A)**. Current is measured using an **ammeter**.

Resistance

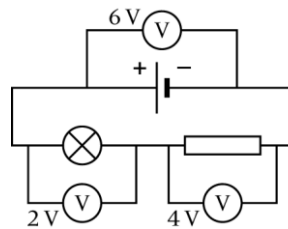
The **resistance** of a circuit is a way of saying how easy or difficult it is for electricity to flow.

- high resistance = hard for electricity to flow = small current
- low resistance = easy for electricity to flow = large current

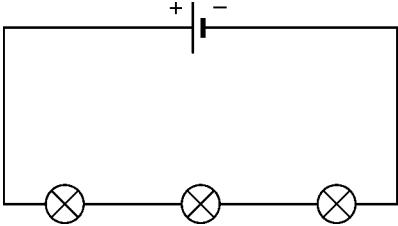
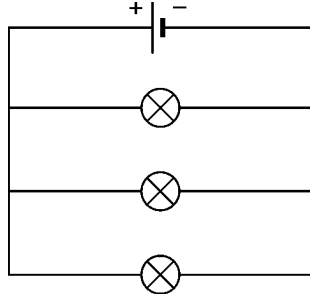
Voltage

A circuit must have a cell or power supply to provide a **voltage**. The voltage pushes the **electrons** around the circuit and gives them energy. This electrical energy is **transferred** to other components in the circuit, which convert it to other forms of energy. For instance, a light bulb transfers electrical energy to heat and light energy.

The voltage of a cell can be measured using a **voltmeter**. The units for voltage are **volts (V)**. The voltage across a component is a way of measuring how much energy the component is transferring. The voltage across all the components in a series circuit adds up to the voltage across the cell.

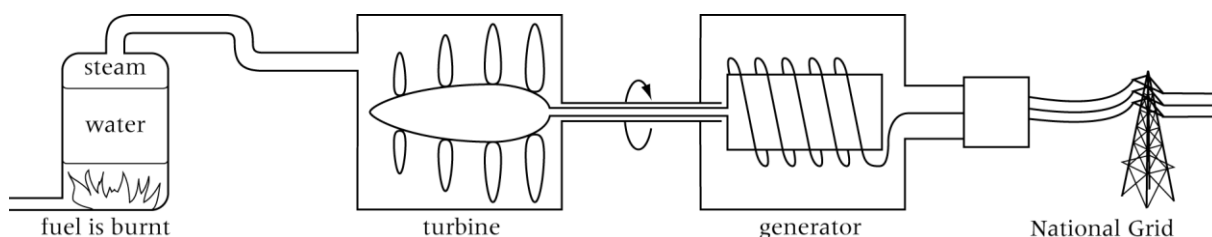


Circuits can be **series** or **parallel** circuits.

Series circuit	Parallel circuit
	
• If one bulb breaks, all the others go off. • The current is the same everywhere. • If you put more bulbs in they will be dimmer, because it is harder for the electricity to get through. The resistance of the circuit is higher. • The voltage from the cell or power pack is divided between the components.	• If one bulb breaks, the bulbs in the other branches stay on. • The current splits up when it comes to a branch. The current in all the branches adds up to the current in the main part of a circuit. • If you add more bulbs they stay bright. It is easier for the current to flow with more branches, because there are more ways for the electrons to go. • The voltage is the same across all the branches of the circuit.

Generating electricity

Fossil fuels are transported to power stations where they are burnt to release heat energy. This heats water, turning it to steam. The steam drives **turbines** which turn **generators**. The electricity generated flows along cables into the National Grid.



Nuclear fuel is made from a radioactive metal called uranium. The energy in nuclear fuel did not come from the Sun.

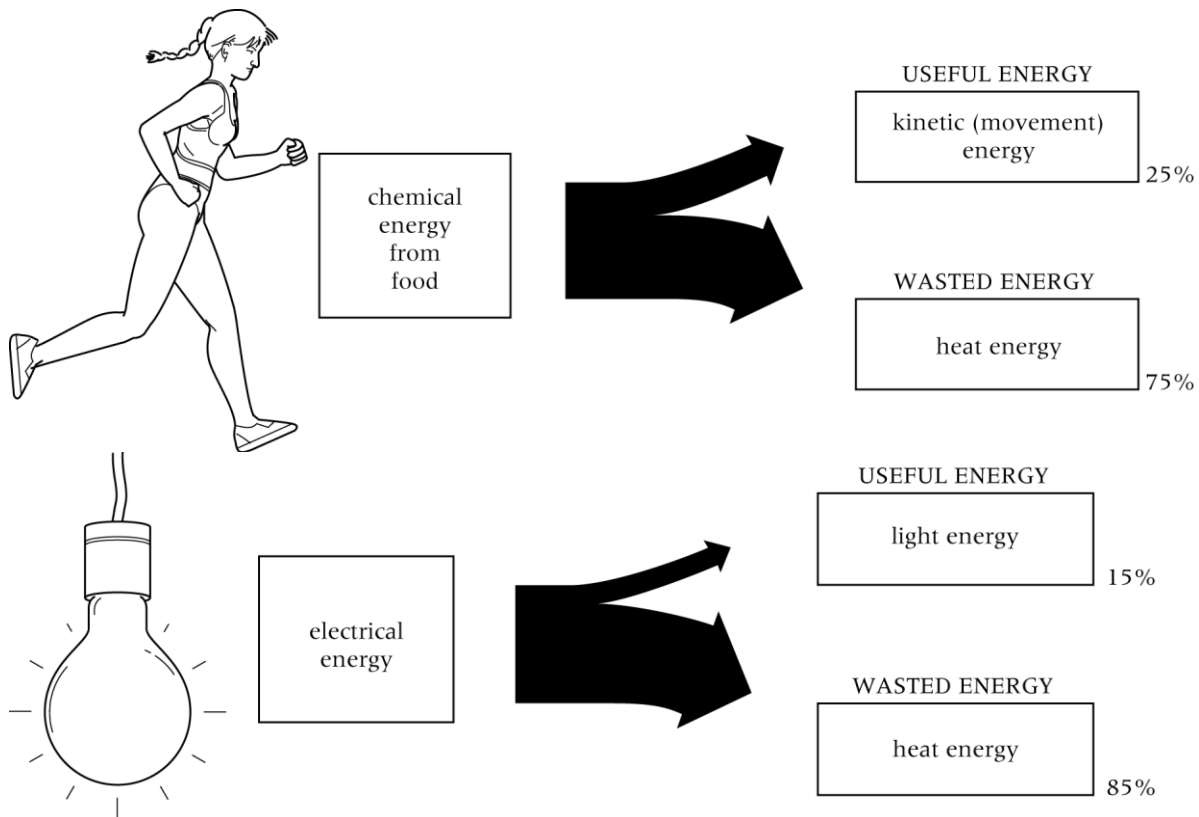
Electricity can be generated from renewable resources such as wind and moving water. These will become more important as fossil fuels run out.

Sometimes we need a source of portable electricity when we are not close to the mains. This can be supplied by **cells** (sometimes called **batteries**). These store chemical energy which can be changed to electrical energy. Cells go flat when they run out of chemical energy. Some cells can be **recharged**.

Wasting energy

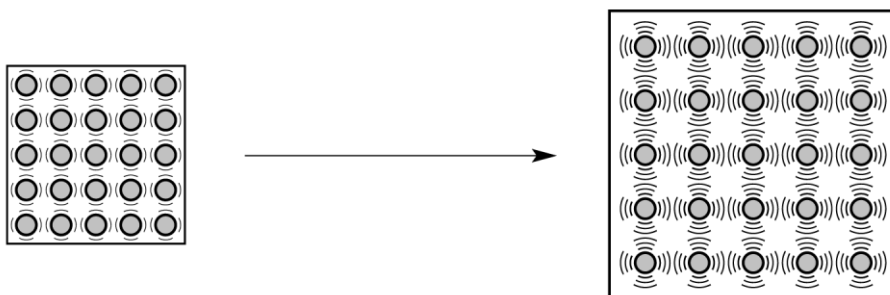
Energy cannot be made or destroyed, but it can be changed to different forms. Not all energy is turned into a form that we want. Often it is turned into heat that we do not need. This is wasted energy. A car engine produces kinetic energy, which is useful. It also produces heat and sound which are wasted forms of energy.

The percentage of useful energy produced by something is known as its **efficiency**. The human body is 25% efficient.



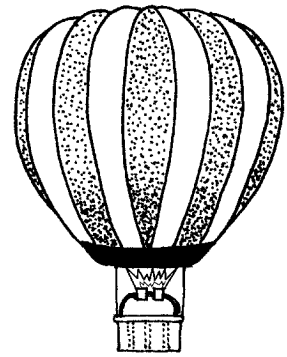
Travelling heat

The **kinetic theory** or **particle model of matter** helps to explain how some forms of heat energy travel. The theory suggests that everything is made of moving or vibrating particles. When these particles are heated they move faster.



When the particles vibrate faster the material expands.

When the air particles in the balloon are heated, they move apart and the air expands and becomes less dense. This causes the hot air to rise, and the balloon rises too.

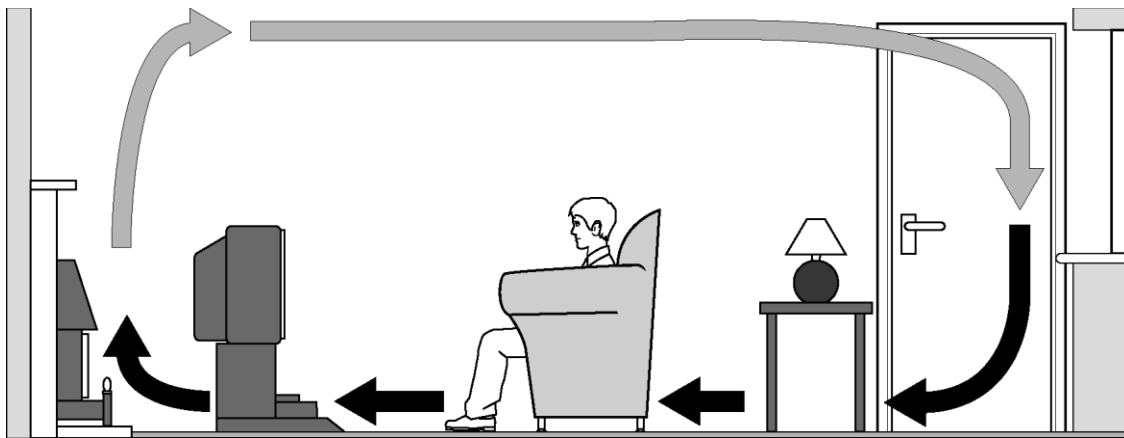


Thermal energy can travel in three different ways.

Conduction takes place in solids and can also happen in liquids (although not very well). The particles in a solid are held together tightly. When they gain energy they vibrate faster and the vibrations are passed on. Particles are not as close in a liquid, so conduction is not very good. Metals are the best conductors. Most other solids are poor conductors.

Something which does not conduct heat very well is an **insulator**. Liquids, gases, and solids which contain a lot of trapped air are insulators.

Convection takes place in liquids and gases.



When the air near the fire is heated, the particles spread further apart and the air becomes less dense and rises. As it rises it meets cooler air and passes the energy on. Having passed on the energy, it cools and becomes denser. The denser air sinks, setting up a cycle or **convection current**.

Heat can be transferred through empty space by **infrared radiation**. Radiation does not require the movement of particles. Any hot or warm object gives off or **emits** radiation. When something takes in heat energy from radiation, it is said to **absorb** it.

Infrared radiation travels as waves. It can be reflected and it can also be focused.



Energy resources

Energy is needed to make things happen. There are different kinds of energy, such as **light energy** and **heat energy** that we get from the Sun, and **electrical energy**.

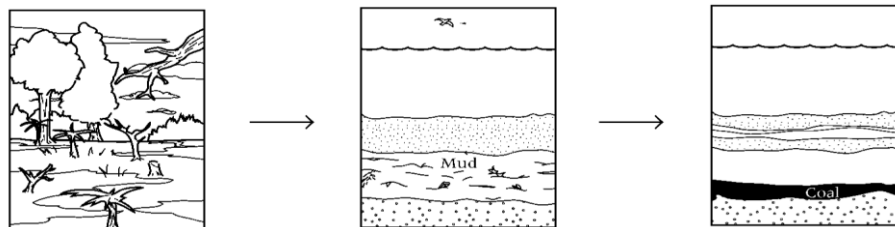
We need **fuels** to provide energy in our homes, factories and for transport. A fuel is something which can release heat energy.

Fossil fuels

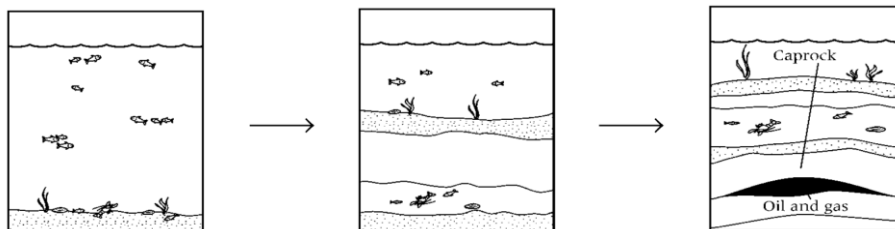
Fossil fuels:

- are made from plants and animals which were trapped in mud and rocks millions of years ago
- include coal, oil and natural gas
- are **non-renewable** (they take millions of years to form, and so our supplies will run out)
- produce gases which cause pollution when they are burnt
- are relatively cheap to obtain
- contain **chemical energy** which changes to heat energy when they are burnt
- originally got their energy from the Sun. The plants that became coal got their energy from the Sun, and the animals that became oil got their energy from plants which got their energy from the Sun.

Electricity is not a fuel. It has to be generated using other energy resources.



How coal is formed.



How oil and natural gas are formed.

Making fossil fuels last longer

We can make fossil fuels last longer by using less energy. We could walk or cycle whenever we can, or use a bus instead of using a car. Walking and cycling would make us fitter and healthier, and there would be less pollution if there were not as many cars on the roads. We could also save energy by keeping our houses cooler and wearing more clothes.

Renewable energy resources

Renewable energy resources:

Biomass – plant or animal material is used to produce heat energy.

Geothermal – hot rocks beneath the Earth's surface provide heat energy.

Solar – light energy from the Sun is absorbed by photovoltaic cells to generate electricity

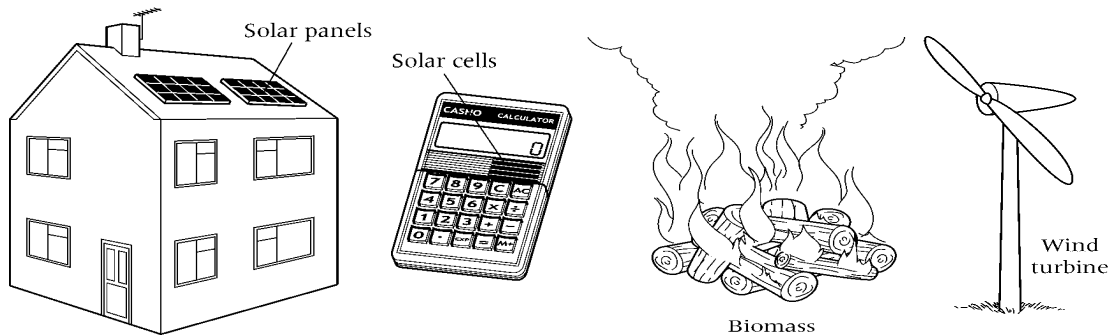
Wind – kinetic energy from the wind is used to generate electricity.

Tidal – kinetic energy from tides moving in and out is converted to electricity.

Hydroelectric – water flows downhill, converting gravitational potential energy into electricity.

Wave - kinetic energy from moving waves generates electricity.

- do not produce harmful gases
- can be expensive
- will not run out.



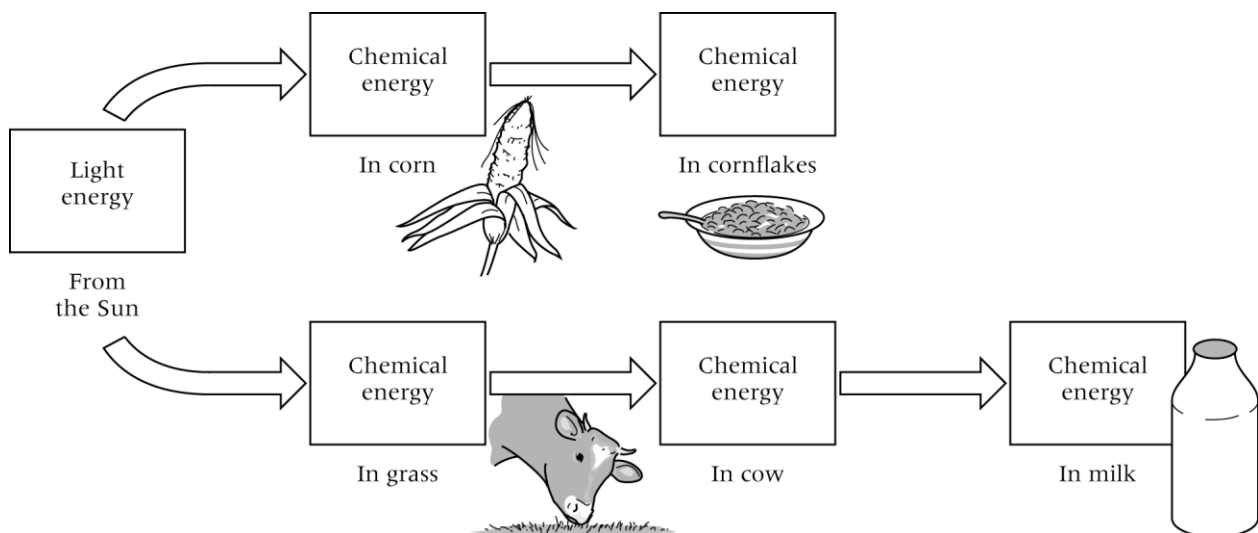
Energy in food

Humans and other animals need energy to live. We get our energy from chemical energy stored in food. We need to choose our food so that we get the right amount of energy. If we eat too much we could get fat and become unhealthy. If we do not eat enough we will get thinner and may become ill.

The unit for measuring energy is the **joule (J)**. There is a lot of energy stored in food, so we usually measure the energy in food using **kilojoules (kJ)**. $1\text{ kJ} = 1000\text{ J}$.

Energy from the Sun

Most of the energy resources we use originally came from the Sun. Only geothermal energy, nuclear power and tidal power do not depend on energy from the Sun.



8P 13b Quick Quiz

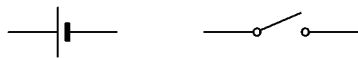
On your answer sheet, write in or circle the correct letter for each question.

1 Which of these statements is true?

- A Plastic is a conductor because it lets electricity flow through it.
- B Plastic is an insulator because it lets electricity flow through it.
- C Metals are conductors because they let electricity flow through them.
- D Metals are insulators because they let electricity flow through them.

The true statement is that energy is conserved: it can be transferred or stored but cannot be created or destroyed.

2 What do these symbols show?



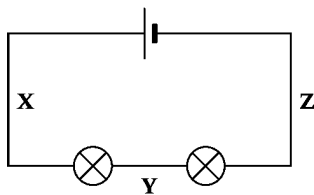
- A a cell and a bulb
- B a cell and a switch
- C a bulb and a switch
- D a switch and a motor

3 Which instrument would you use to measure current?

- A an ammeter
- B a cell
- C a motor
- D a voltmeter

4 How big would the current be at X, Y and Z?

The current is the same at X, Y and Z because they are all in the same series circuit - current is not used up as it flows round the loop.



- A It would be biggest at X and smallest at Z.
- B It would be biggest at Z and smallest at X.
- C There would be no current at Y.
- D It would be the same at all three places.

5 You can make the current in a circuit smaller by:

- A putting a switch in the circuit.
- B putting more bulbs in the circuit.
- C taking some bulbs out of the circuit.
- D putting more cells in the circuit.

6 Resistance is a way of saying:

- A how much electricity is flowing.
- B how hard it is for electricity to flow
- C how many cells are in a circuit.
- D how many switches are in a circuit.

7 Electricity is:

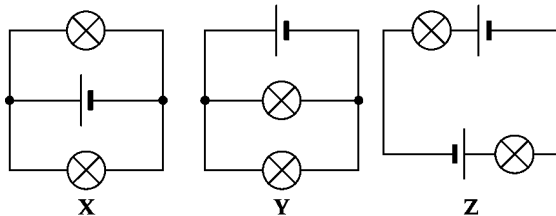
- A a source of energy.
- B a kind of liquid inside the wires.
- C tiny particles called electrons flowing through the wires
- D tiny particles called atoms flowing through the wires.

8 A long wire:

- A has a smaller resistance than a short wire.
- B has the same resistance as a short wire.
- C has a bigger resistance than a short wire
- D does not have resistance.

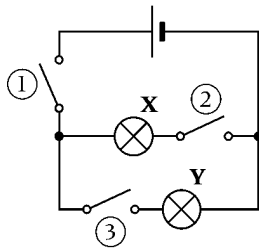
9 Which of these circuits are parallel circuits?

The parallel circuits: components connected in separate branches across the supply, so the current splits between branches and each branch gets the full supply voltage.



- A X and Y only
- B X and Z only
- C Y and Z only
- D all of them

10 Which switches must be pressed to make bulb X come on?

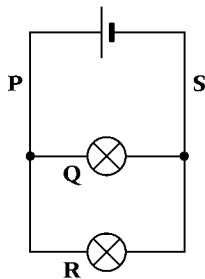


- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D all three of them

11 Parallel circuits are used for house lights:

- A so they can all be switched on and off together.
- B so you know when one breaks, because all the others go off.
- C so they use a bigger current.
- D so they can all be switched on and off separately.

12 Which of these statements is **not** true?



- A The currents at Q and R add up to give the current at S.
- B The current is the same at P and S.
- C The current is the same at Q and R.
- D The current is the same at P and Q.

13 Which of these is **not** a form of energy?

- | | |
|-----------|----------|
| A kinetic | B petrol |
| C light | D sound |

14 Which of these has the most gravitational potential energy?

- A a parachutist standing on a runway
- B a parachutist going into a plane
- C a parachutist jumping out of a plane
- D a parachutist after landing

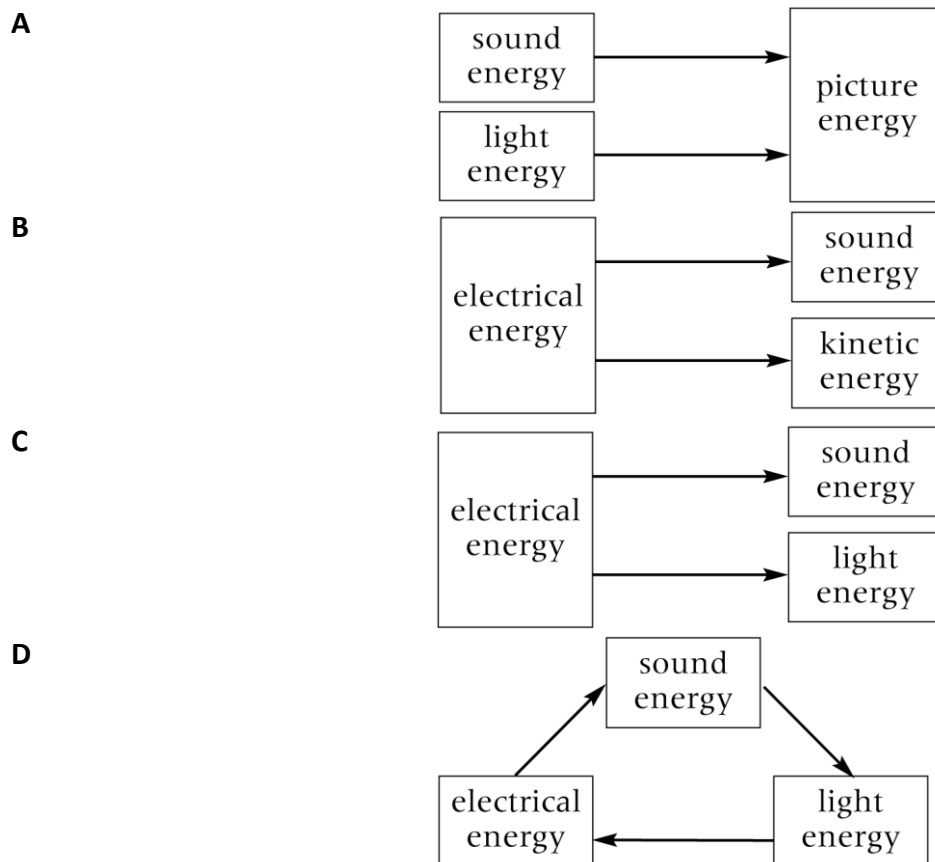
The object with the greatest mass at the greatest height · gravitational potential energy = mass × g × height, so biggest mass × height wins.

15 What are the energy changes in a candle?

- A heat to light
- B heat to sound
- C light to heat
- D chemical to heat and light

Chemical energy stored in the wax is transferred to light energy and thermal (heat) energy.

16 Which is the correct energy flow diagram for a television?



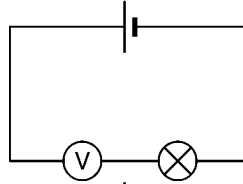
· Correct diagram: electrical energy in, light and sound energy out (heat also wasted). Choose the option showing electrical energy → light + sound energy.

17 The units for voltage are:

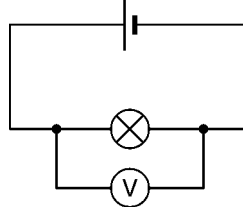
- A amps.
- B ohms.
- C volts
- D metres.

18 Voltage is measured:

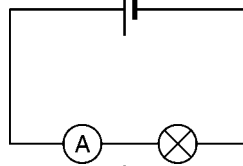
A using a voltmeter in the circuit like this.



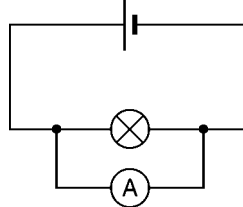
B using a voltmeter in the circuit like this.



C using an ammeter in the circuit like this.



D using an ammeter in the circuit like this.



19 Voltage is:

A another name for current.

B a way of counting how many cells there are in a circuit.

C a way of saying how much energy the electricity is carrying.

D something that makes it difficult for electricity to flow.

20 Which statement is **not** true for a series circuit?

A The voltage is always the same for all the components.

B The voltage across each component depends on how much energy it uses.

C The voltage across all the components adds up to the voltage across the cell.

D The voltage can be measured using a voltmeter.

21 Which is true? *In a series circuit the current is the same at every point and the supply voltage is shared between the components.*

A Renewable energy resources will run out one day.

B Renewable energy resources will not run out.

C Non-renewable energy resources will not run out.

D None of our energy resources will run out.

22 Which of these is a non-renewable energy resource?

- A solar
- B wind
- C natural gas
- D moving water

Coal (also oil, natural gas or uranium) · fossil and nuclear fuels are non-renewable; wind, solar, hydro and tidal are renewable.

23 Which statement is true?

Renewable resources such as moving water, wind and solar are replaced as they are used, so they will not run out.

- A Solar panels are used to generate electricity.
- B Solar cells and solar panels are the same.
- C Solar cells are used to generate electricity.
- D Solar cells are used to heat water.

24 Biomass is:

- A a heavy weight.
- B any fuel made from plant or animal material
- C a fuel made from plant material only.
- D a fuel made from animal material only.

25 Which of these materials is a good conductor of heat?

- | | |
|-----------------|---------|
| A <u>copper</u> | B air |
| C water | D glass |

26 When one end of a piece of metal is heated:

- A the particles get closer together.
- B the particles vibrate more and the vibrations are passed from particle to particle.
- C the heat does not move through the metal.
- D the particles get bigger.

27 Solids are better conductors than liquids or gases because:

- A the particles are closer together in gases.
- B the particles are closer together in liquids.
- C the particles are closer together in solids.
- D the particles in a solid cannot move.

28 Feathers are good insulators because:

- A they are solid.
- B birds have them.
- C they trap pockets of air
- D they are waterproof.

29 Why do materials expand when they are heated?

- A The particles get bigger.
- B The particles move around more and take up more space.
- C The particles get closer together.
- D The particles stick together.

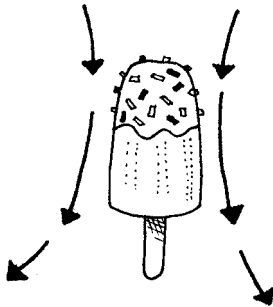
The particles gain kinetic energy and vibrate faster, so the average spacing between them increases and the material takes up more space.

30 Which substances can heat travel through by convection?

- A solids and liquids
- B solids and gases
- C only liquids
- D liquids and gases

Liquids and gases (fluids) only · their particles can move freely and form convection currents; solids cannot convect.

31 Why does the cold air near the ice lolly move downwards?



- A It is colder and denser than the surrounding air.
- B It is warmer and denser than the surrounding air.
- C It is colder and less dense than the surrounding air.
- D It is warmer and less dense than the surrounding air.

32 Infrared radiation can travel through:

- A transparent objects only.
- B opaque objects only.
- C empty space only.
- D empty space and transparent objects.

33 Which forms of energy are often produced as wasted energy?

- A heat and light
- B heat and sound
- C sound and light
- D sound and kinetic

34 Kinetic Energy is:

- A** Measured in Newtons
- B** A type of fuel
- C** A form of energy in moving things
- D** Stored energy

8P 13c Reflection on Inquiry

Aim: Reflect on the Knowledge and Skills acquired in the Energy Production and Climate Change Unit through Inquiry

Questions asked	Answers found, and how we found them	Any further questions?
Factual - What technologies are available to produce sufficient energy to meet the needs of the human population? - What are the advantages and disadvantages of different sources of energy?	We researched online and in class and found: renewables (wind, solar, hydro), nuclear power and fossil fuels. We compared them using tables and class discussion.	Which mix of sources would be cheapest and most reliable for our town?
Conceptual - How is electrical energy made and transferred? - How will the continued use of fossil fuels affect the environment and humankind? - How can the different sources of energy production be best utilised to minimise environmental change? Electrical energy is made by a generator turned by steam (from burning fuel or nuclear fission) or directly by wind/water: it is transferred by cables and pylons to homes and industry. Use renewable sources (wind, solar, hydro, tidal), improve efficiency to reduce wasted energy, and cut fossil fuel use to	We traced energy chains (fuel · heat · kinetic · electrical) and used generators and circuits to see how kinetic energy becomes electricity.	How can electricity be stored for when the wind does not blow?
Debatable - Is global collaboration required to minimise or prevent the consequences of climate change? - Is the use of nuclear power safe?	We debated fossil fuels: burning them releases CO₂ which causes global warming, while renewables and nuclear release little or none. We weighed cost, reliability, safety and pollution. Science cannot fully prevent the consequences, but it can reduce them (renewables, carbon capture) and help us adapt (flood defences, drought-resistant crops).	Is it fair to expect poorer countries to cut emissions too? Should we build more nuclear power stations?
Reflection on ALT (approaches to learning) used: - Communication - Collaboration - Organisation - Research - Critical Thinking - Creative Thinking	Communication: debated in role; Collaboration: group work; Organisation: planned research; Research: gathered facts; Critical Thinking: weighed opinions; Creative Thinking: suggested solutions.	I used teamwork and research most. Next time I will record my sources and listen to other views before deciding.

8P 15a Welcome to Nimby

Newspaper Extract:

**What kind of Power - Nimby must decide.
Village meeting next week.**

Lynne Clarke reports.

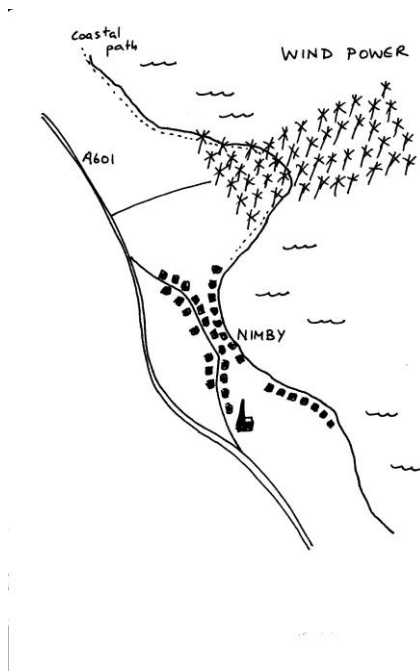
Next week the residents of Nimby will be consulted about how energy is provided for their growing population. The old coal power station at Nimby Ness is to close down and be dismantled. Two plans have been put to Nimby Council for a new way of providing energy. The first is for a nuclear power station to be built on the site of the old coal station. The second is for a wind farm of 100 wind turbines to be placed on Nimby Ness, extending 5km out to sea. The location at Nimby Ness is ideal for power generation as there is a water supply which could be used for cooling in the nuclear plant, as well as adequate wind speed from shoreline breezes.

A nuclear power station would also be able to supply energy to Benwick, the nearest small city. The wind generators would be able to supply a number of small towns like Nimby, but only part of the city.

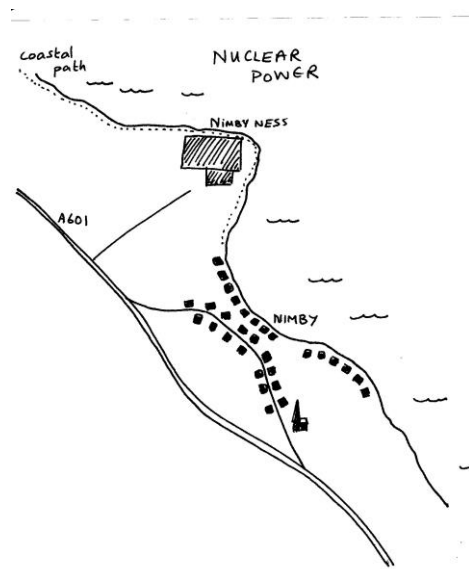
While the new installation is being built, which is estimated to take around 5 years, Nimby will import energy from Norway along cables laid under the North Sea. This is an expensive way of buying energy and isn't affordable in the long term. If the wind farm is chosen, there will still be a need for Nimby to take energy from other sources across the country, including nuclear stations.

At the crucial meeting, which will be held in Nimby village hall, representatives from Scottish Nuclear Power and Northern Wind will each present their case. Farmers and shopkeepers will be present alongside village residents and local councilors, who will have to take the final decision on the type of power station to be built. Planned locations for the rival stations are outlined on the next page.

Wind Power Solution



Nuclear Power Solution



Energy and Nimby – your views for the future.

We will be joined by experts on nuclear power and wind power. You will have a chance to say what you think and listen to others. We need to make decisions on how to supply the extra energy that Nimby needs. Your voice counts.

Questions for you to address:

- What role will I play? How can I get into the role?
- What do I need to find out in advance about nuclear power and about wind power?
- What might the perspectives of the other represented groups be?
- What questions might I need to prepare in advance – what would I like to know?

Choose a role to represent (e.g. nuclear company spokesperson, local resident, wildlife trust, government minister) and prepare arguments from that viewpoint.

Nuclear: how fission works, cost, radioactive waste, accident risk and CO₂ emissions. Wind: output, cost per kWh, land/sea space, noise, visual impact and intermittency.

Residents may worry about noise and views; wildlife groups about birds and habitats; energy companies about cost and profit; government about energy security and emissions.

Prepare questions on cost per kWh, capacity and reliability, waste disposal and safety, carbon emissions and land use so you can challenge the other groups.

8P 15b Meeting Planning

You will be allocated one of the following roles:

- Nuclear Power Expert
- Wind Power Expert
- Farmer
- Local Councilor
- Shopkeeper
- Villager



You must carefully read the information about your role. Pay close attention to what you need to do to prepare for the meeting.

Role:

My role in the meeting is to represent my allocated role and its viewpoint (e.g. Nuclear Power Expert)

How can I get into the role?

I will read my role card carefully, research nuclear and wind power, and prepare arguments to present at the village meeting.

I will practise speaking in role and prepare answers to the questions other groups might ask me.

Research:

What do I need to find out in advance about nuclear power and wind power?

Here is some space for your research:

~~Research: nuclear - no CO2 but radioactive waste and accident risk; wind - renewable and clean but unreliable; Nimby
Ness has good wind and cooling water.~~

Perspectives:

What might the perspectives of the other represented groups be? Try to come up with counterarguments if they agree with your views, or supporting arguments if they agree.

Farmers worry about turbines on farmland and the view; shopkeepers want cheap, reliable power; the council wants the safest, cheapest option; villagers worry about safety; experts support their own technology.

Questions:

What questions might I need to prepare in advance – what would I like to know?

Who will you ask these questions to? I will ask the nuclear and wind power experts, then the council and villagers for their views.

How safe is nuclear power and what happens to the waste? How much electricity will the wind farm make, and what happens when the wind drops? How much will each option cost?

Other Considerations / Additional Notes:

~~Keep an open mind: listen to all groups and base my final decision on evidence about cost, reliability, safety and the environment.~~

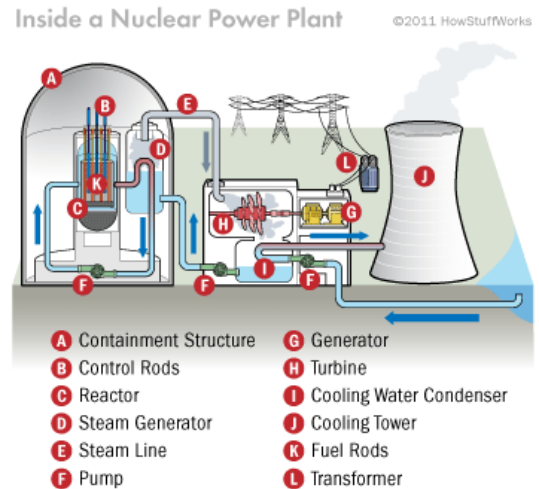
8P 16a Nuclear Power

How does it work?

Nuclear power stations use the energy released during **nuclear fission** reactions. The most common fuel used is uranium.

There are 3 parts of a nuclear power station:

1. Reactor
2. Turbine
3. Generator



In the table below, identify the energy transformations that take place in each part.

Nuclear fission of uranium releases heat energy; the heat boils water into steam, which turns a turbine driving a generator, producing electrical energy.

Part	Energy Transformation
reactor	nuclear energy → heat energy
turbine	heat energy → kinetic energy
generator	kinetic → electrical

Advantages and Disadvantages:

Write down 3 advantages and 3 disadvantages of using nuclear power.

Advantages:

1. Produces no greenhouse gases or air pollution while generating electricity
2. Generates a huge amount of energy from a very small amount of fuel
3. Very reliable · it can run continuously for long periods of time

Disdvantages:

1. Produces radioactive waste that remains dangerous for thousands of years
2. Risk of nuclear accidents which can release harmful radiation
3. Very expensive to build, maintain and decommission

So.....

What is your opinion on Nuclear Power?

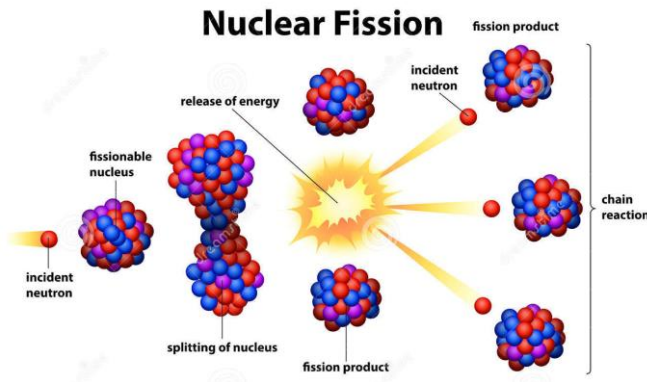
Nuclear power produces large amounts of reliable, low-carbon energy, but it creates radioactive waste and carries accident risk; I think it is useful if managed very safely.

8P 16b Nuclear Power Extension Work

Nuclear Fission

A nuclear reactor uses a nuclear fission reaction to produce energy. Here is a diagram to illustrate a nuclear fission chain reaction:

Use this diagram to explain what happened in nuclear fission reaction:



A neutron is fired at a uranium-235 nucleus.

The uranium nucleus absorbs the neutron and becomes unstable.

The nucleus splits into two smaller nuclei, releasing energy as heat.

Two or three more neutrons are also released.

These neutrons strike other uranium nuclei, which split too.

Each split releases more neutrons, so the reaction carries on.

This is a chain reaction, releasing a huge amount of energy

Control rods absorb neutrons to keep the reaction at a safe, steady rate

Parts of a Nuclear Power Station

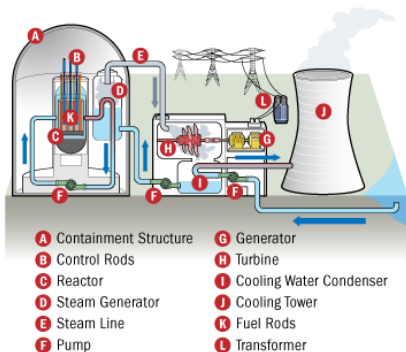
You should know that a nuclear power station is comprised of 3 parts:

1. Reactor
2. Turbine
3. Generator

You should also know the energy transformations that take place in each part. There are however more parts that play an important role in ensuring safety and efficiency.

For each part listed in the table, research what it does then describe its function:

Inside a Nuclear Power Plant © 2011 HowStuffWorks



Part	Function
Containment Structure	A strong concrete and steel building that contains the reactor to stop radiation escaping
Control Rods	
Pump	
Condenser	
Cooling Tower	Releases waste heat from the cooling water into the atmosphere
Steam Line	
Transformer	
Fuel Rods	
Moderator	Slows down the neutrons so they are more likely to cause further fission

8P 16c Wind Power

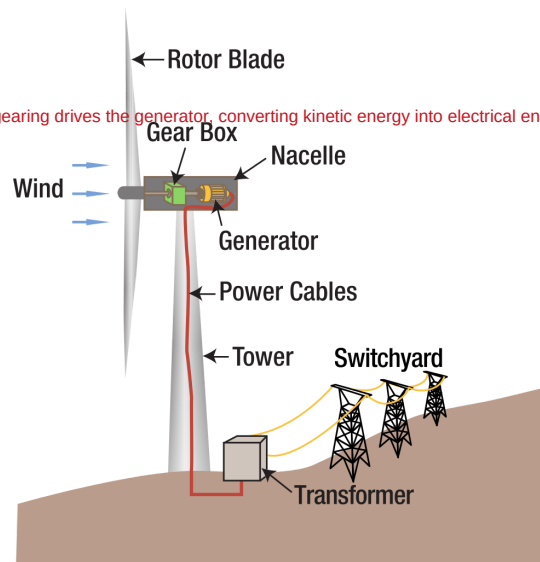
How does it work? The kinetic energy of the moving air turns the rotor blades; gearing drives the generator, converting kinetic energy into electrical energy.

The kinetic energy of the wind turns the rotor. This then spins the dynamo which converts the KE into electrical energy.

There are 3 parts of a wind power turbine:

1. Rotor
2. Generator
3. Gearing systems

In the table below, identify the energy transformations that take place in each part.



Part	Energy Transformation
rotor	kinetic energy from wind → rotational kinetic energy
dynamo or electrical generator	kinetic energy → electrical energy

The gearing systems ensure that the turbine is always facing into the wind to maximize the available wind energy

Advantages and Disadvantages:

Write down 3 advantages and 3 disadvantages of using wind power.

Advantages:

1.

2.

3.

Renewable · the wind will not run out

Disadvantages:

Produces no greenhouse gases or pollution while generating electricity

1.

2. Can be noisy and spoil the landscape; can harm birds

3. Expensive to build and needs backup power for calm days

Cheap to run once the turbines have been built

So.....

What is your opinion on Wind Power?

Wind power is renewable and produces no greenhouse gases, but it is unreliable when the wind drops and can spoil the landscape, so it works best alongside other sources.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p14: Sankey diagram: one arrow 10 squares wide going in (INPUT ENERGY 10 J). Split into a 4-square-wide arrow (USEFUL OUTPUT ENERGY - light, 4 J) and a 6-square-wide arrow (WASTED OUTPUT ENERGY - heat, 6 J).

p29: Circuit B diagram: draw the cells in series with correct symbols, making sure every cell points the same way round, joined to the bulb in one loop.

p29: Circuit C diagram: draw the same circuit but ensure both cells point the same way round (the long line/positive terminal of each cell facing the same direction).

p58: Boiler - where the fuel burns and heats the water. Turbine - spun around by the high-pressure steam. Generator - turned by the turbine to produce electricity. Transformer - steps the voltage up for transmission. Cooler - condenses the used steam back into water.

p70: Problems: CO₂ emissions cause climate change; sulphur dioxide causes acid rain; soot causes air pollution and health problems; oil spills and mining damage habitats; fossil fuels are non-renewable.

p72: Compare each option: how many homes it supplies, its cost per unit of electricity, and the site requirements (wind speed, estuary/tidal range, wave exposure, sunlight, river) shown on the map.

p72: Visual impact · turbines or pylons on a hill can spoil the view for local people and tourists; offshore or screened sites reduce this.

p72: Natural heritage · could harm plants and animals, e.g. birds and bats killed by turbine blades or fish blocked by a hydro scheme; surveys and mitigation reduce this.

p72: Model justification: choose offshore wind plus solar - the coast is windy, so wind gives the most clean electricity at the lowest running cost. Reject a tidal barrage (very costly, few suitable estuaries, harms wildlife).

p100: Absorbs neutrons to control the rate of the fission reaction

p100: Circulates the coolant (water) around the reactor to carry heat away

p100: Cools the steam back into water so it can be reused in the system

p100: Carries high-pressure steam from the reactor to the turbine

p100: Steps up the voltage so electricity can be sent through the National Grid

p100: Contain the uranium fuel in which the fission reactions take place

p101: kinetic energy from wind · rotational kinetic energy